

Data Driven Approach for Root Cause Identification

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1 Introduction

Petrochemical plants (ethylene/propylene manufacturing units) produce ethylene and propylene by cracking saturated hydrocarbon feedstock (ethane and propane) in a pyrolysis furnace. Acetylene is formed as a by-product in these reactions. This acetylene impurity in the ethylene product affects its polymerization process, as small amounts of acetylene can impart undesired properties to the polymer products. To prevent this, acetylene concentration is maintained at less than 1 ppm. This is achieved by selectively hydrogenating acetylene to ethylene, which brings the added advantage of improving the plant's overall ethylene yield.

The system under consideration, a tail-end acetylene hydrogenation reactor, in which hydrogen is injected into the feed of the reactor, is used to prepare feed for further processing. This configuration is less demanding on the equipment, catalyst, and operation and gives better ethylene yield over the long term. Two such reactors are typically used in the set-up, with one reactor in the regeneration or standby mode while the other is online. Depending on vendor specifications, reactor online cycle times typically span five to six months. The cycle time and reactor configuration can vary based on the acetylene load at the inlet.

The catalyst used in the system is Pd-Ag/Al₂O₃. Over time, the catalyst activity decreases due to catalyst fouling, poisoning, sintering, and improper or frequent regeneration. Additionally, the catalyst may lose its selectivity concerning the formation of ethylene. The reactor system also employs CO to temporarily poison the catalyst and inhibit side reactions, i.e., the hydrogenation of ethylene to form ethane. The mechanism by which this occurs is provided in [9]. CO is usually injected when the catalyst is fresh, or to shift the reaction from the top section to the mid section of the bed.

Another undesired side reaction is the oligomerization of acetylene to form green oil, which deposits on the surface of the catalyst and brings down its activity. The equations in [1] model this deactivation. The decrease in activity is compensated by increasing the reactor temperature; however, this can further accelerate catalyst deactivation, and cause an increase in ethane production. Eventually, the reactor cycle cut-off/ catalyst changeover criteria is determined by outlet temperature (100-110 ° C), the selectivity of the catalyst (EOR: 10%), and the outlet acetylene concentration (1 ppm). The objective of this work is given below:

1.1 Objective of the work

- To Find the root cause for the slippage event (acetylene (C_2H_2) above (1ppm) with a data-driven method.
- The data-driven method should be explainable.

1.2 Organisation of the report

This report is structured into Six sections. The **First** section provides an introduction, followed by Section **Two**, which presents a literature survey. Section **Three** outlines the modeling and root cause identification approach implemented to achieve the objectives. Section **Four** is the implementation of the discussed method on a toy example. Section **Five** discusses data preprocessing. Section **Six** covers the implementation of the described methodology to the dataset provided by Ingenero. Finally, Section **Seven** presents the conclusion.

2 Literature Survey

Having comprehensive fault detection and isolation (FDI) strategies in a chemical process is important to ensure safe production, reduce variability of products and reduce the manufacturing cost [40]. In literature, various available FDI approaches can be broadly classified in three categories: (1) Knowledge based, (2) Model based, and (3) Data driven approaches. The knowledge based and model based approaches require detailed understanding or first-principles model of process to be able to efficiently monitor the process. However, developing an accurate model is generally very difficult, time consuming, and expensive, specially for large scale systems. In contrast, the data driven approaches aim to extract relevant model from available data, and are generally comparatively easier to apply and computationally inexpensive for FDI problem. However, to be effective, they need access to large amount of good quality data which should include measurements of all the key variables in a process.

As part of the current work, a comprehensive literature survey of data driven approaches was performed to understand various methods for FDI and root cause identification. A summary of the findings is provided in Table 1. Key idea in many data-driven approaches is to characterize normal process operation by some statistical model. Correspondingly, a relevant statistic is monitored. Violation of pre-specified threshold by the statistic indicates occurrence of a fault. Subsequently, root cause or variables contributing to the fault are identified using a variety of approaches. Amongst the various approaches used for FDI as listed in Table 1, some popularly used data driven modeling ideas are shortlisted and highlighted next:

1. Shewhart charts [37] for univariate statistical process control, and Hotelling T^2 control charts [4] for multivariate statistical process control. These models detect out-of-control operation (or a fault). Subsequently, T^2 decomposition based approach can be used for identifying variables contributing to the fault.
2. Projection methods such as Principal Component Analysis (PCA) [2], Independent Component Analysis (ICA) [29] and Partial Least Square (PLS) [18]. These methods are used to extract the key features of the given data, and these key features are used to compute the contribution of each variable when a fault event occurs.
3. Information theory based methods including Information entropy (IE) ([8], [31]), and Mutual information (MI) [24]. These methods use ideas from information theory to detect faults, and to establish root causes.

For the specific problem of root cause identification, apart from the above approaches, other approaches which explicitly incorporate time-series nature of data, such as Granger causality [14] have also been proposed.

In subsequent sections, we survey the Hotelling- T^2 , and information theory based approaches in some detail.

2.1 Causation based T^2 decomposition method [40]

The Hotelling T^2 control charts provide an effective way to detect a fault in a process. When all variables are considered directly in T^2 computation, it boils down to Mahalanobis distance. In particular, T^2 value at any instant in time is a measure of distance between measurements at that time from the mean values corresponding to normal process operation. Unlike conventional distance, T^2 takes the correlation across variables into account. While a large value of T^2 indicates a fault or that the process is out of control, on its own it does not provide the root cause for the fault event [4]. Mason et al. [33] modified the Hotelling T^2 to enhance its diagnosing ability. In particular, they proposed decomposing T^2 into a set of independent parts to facilitate its interpretation. However, the approach was computationally expensive as the number of possibilities increases rapidly with the number of variables. Li et al. [30] proposed causation based T^2 decomposition method for FDI problem to overcome this issue. Their method is able to reduce the computational time and improve the diagnosability of T^2 approach. Verron et al. [40] combined the approach of Li et al. [30] on causal decomposition of T^2 with the use of Bayesian network [41] for root cause identification. In particular, in their work [40], T^2 decomposition is performed based on Bayesian network and corresponding decomposed T^2 terms are used for fault isolation. Use of this approach requires knowledge of a causal network and prior probabilities of fault events. For a high-dimensional illustrative problem, the causal network was obtained using a data-driven approach, namely the Peter and Clark (PC) algorithm [27]. Verron et al. [40] demonstrated the utility of the proposed method by applying it to two simulation based problems: (i) Hot forming process, and (ii) Tennessee Eastman process (TE). These two applications are briefly discussed next.

Problem 1: Hot forming (HF) simulation process [40]

The HF process depicted in Figure (1) has five variables: One is a quality variable (X5: final dimension of workpiece), with remaining four being process variables (X1: temperature, X2: material flow stress, X3: tension in workpiece, and X4: blank holding force). The causal corresponding Bayesian network represented as a digraph is presented in Fig. 2. The T^2 decomposition based

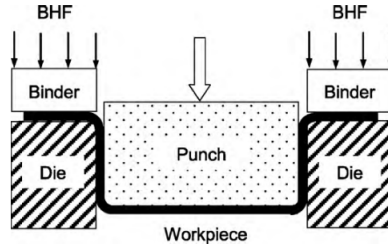


Figure 1: Hot forming process. (Reproduced from [40])

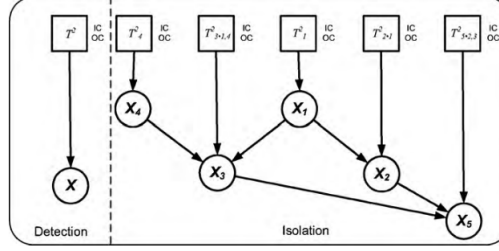


Figure 2: Hot forming: Digraph. (Reproduced from [40])

on this digraph is:

$$T^2 = T_1^2 + T_4^2 + T_{2|1}^2 + T_{3|(1,4)}^2 + T_{5|(2,3)}^2 \quad (1)$$

In the above expression $T_{X_i|(X_{i_1}, X_{i_2}, \dots, X_{i_p})}^2$ indicates T^2 of variable X_i after it is conditioned on variables $X_{i_1}, X_{i_2}, \dots, X_{i_p}$. Note that the way of conditioning variables in Eq. (1) is consistent with the process digraph as represented in Figure 2. To showcase their approach, [40] performed simulations. In particular, first 50 observations are generated corresponding to normal operation (In Control situation) while the next 50 observations correspond to occurrence of faults (Out of Control situations). For fault conditions, step up signal is given to variables X_1 and X_4 . The author assumed 0.5 as a-priori probability of the faults. For each variable, the out control situation is then investigated on the basis of priori and posteriori probabilities. If the posteriori probability becomes higher than the priori probability for a given variable, it can be considered to be out of control i.e. as a root cause variable. The authors compare their approach with classical approach (Shewhart control chart). The T^2 decomposition plot is presented in Fig.3 where first and second columns represent the classical and proposed approaches respectively. From Fig. 3 the following can be noted: (i) Classical approach: Even though variables X_1 and X_4 are the root causes (after 50 instants), all the individual variables and the overall T^2 statistic exceed their control limits (indicated by red lines). Thus while the classical approach indicates presence of a fault after 50 instants since T^2 is high, it does not identify the root cause. (ii) Bayesian network based method: the plots in the second column correspond to this. The y-axis labels in plots in this column indicate the probability of the corresponding event (variable or process being out of control (OC)). For example, the plot in top panel second column gives the probability of the process being out of control. It can be observed that the process is identified to be faulty after 50 instants (top panel). Further, only variables X_1 and X_4 are identified as root cause of the fault event because only for these two variables the conditional posteriori probability crosses the a priori probability (the red line corresponding to probability of 0.5).

Problem 2: Tennessee Eastman (TE) simulation process [40]

The Tennessee Eastman problem is a benchmark simulation problem available

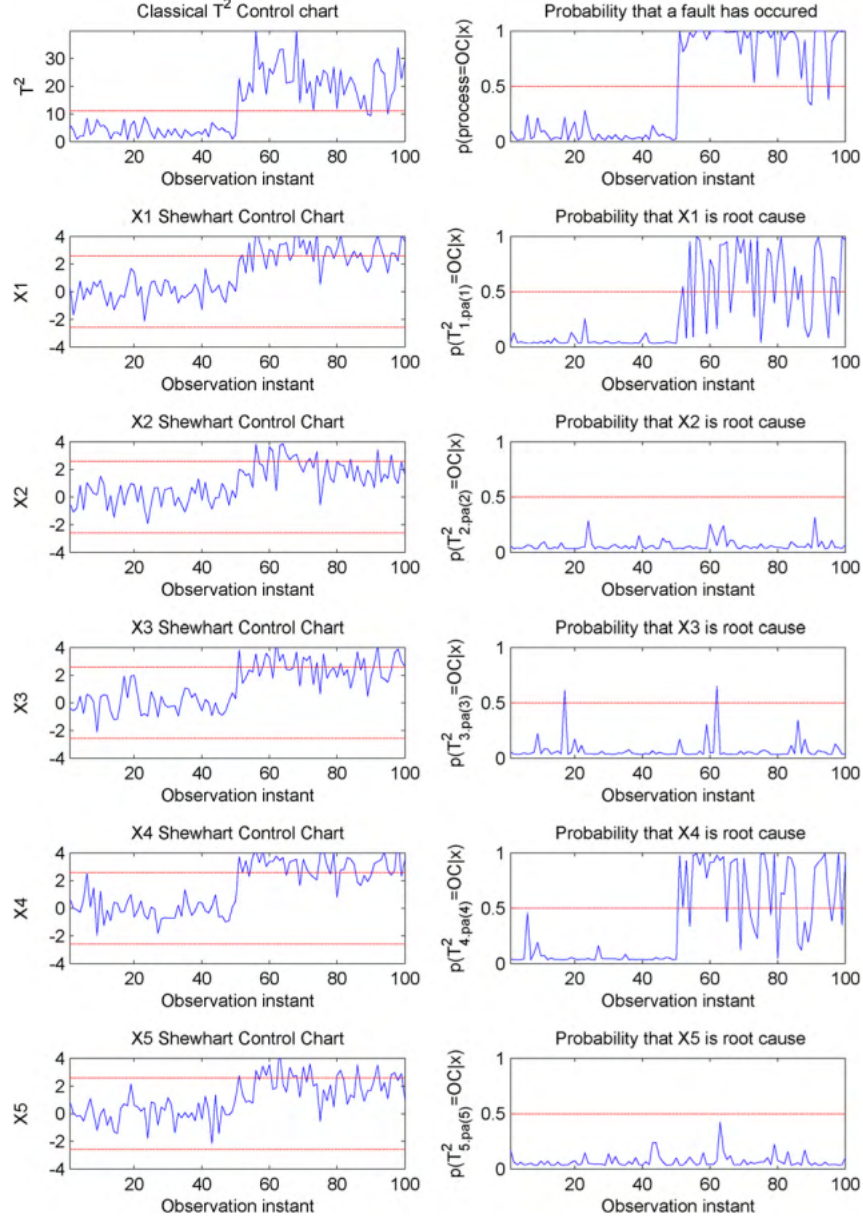


Figure 3: Hot forming: T^2 decomposition. (Reproduced from [40])

in FDI literature to enable checking of the effectiveness of various diagnostic methods and has been widely used. The process flowsheet is presented in Fig. 4. It has four main units: reactor, condenser, separator and stripper. Even though

it is a simulation study, the process flowsheet mimicks an industrial process with large number of states (51), nonlinearities, and coupled nature (with recycle) of the process [7]. Additionally, several faults can be simulated in the process. In their work [40], the authors listed the results for fault-4 to check the efficacy of their Bayesian network based FDI approach. This fault corresponds to a step change in the reactor cooling water inlet temperature. The inlet temperature is not measured but this fault can be observed by values of other two variables: Reactor temperature (X9) and Reactor cooling water flow (X51). In the work of [40], the causal network of TE process is obtained using the PC algorithm and the corresponding digraph is presented in Fig.5. For digraph construction using PC algorithm, 800 fault free observations were considered. Here, PC algorithm used the partial correlations for evaluation of conditional independence. To test the performance 800 observations corresponding to fault 4 were simulated. The proposed method was able to detect 771 observations as faulty (accuracy of 96.38%). Moreover, the root cause of faults were identified to be variables X9 and X51 as represented in Fig. 6 for approximately 900 sample point. From this figure it can be observed that around 190 instant the reactor temperature (X9) shot up due to the fault. As a result, due to controller action the flowrate of cooling water (X51) also increased to maintain constant reactor temperature (X9).

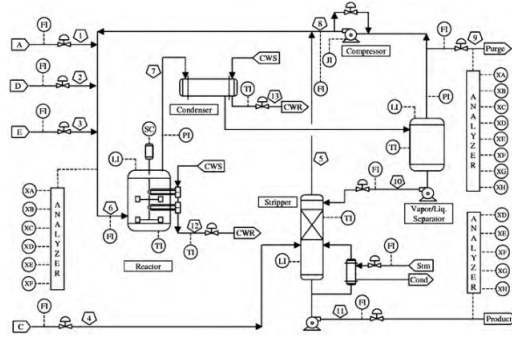


Figure 4: Tennessee Eastman: Process flow sheet (Reproduced from [40])

The above case studies demonstrated the efficacy of the causal network based T^2 -decomposition approach to detect faults and identify the root causes. However, T^2 decomposition method in its native form has limitations in that it assumes the following:

1. Variables follow Gaussian distribution
2. Variables are linearly related
3. Only acyclic digraph is considered for T^2 decomposition
4. The process is stationary in nature

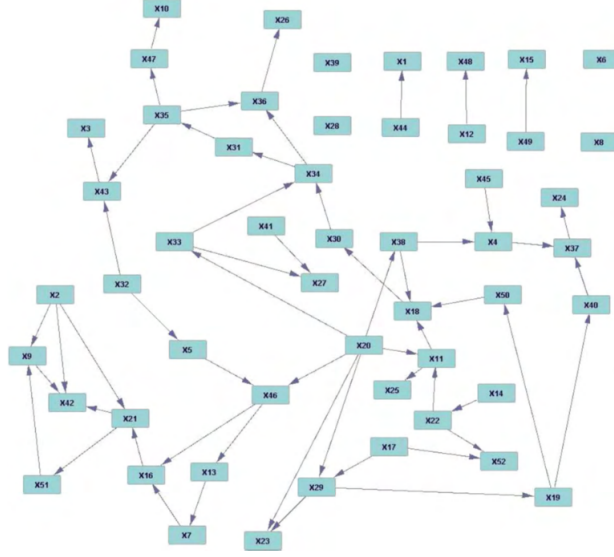


Figure 5: Tennessee Eastman: Digraph (PC algo). (Reproduced from [40])

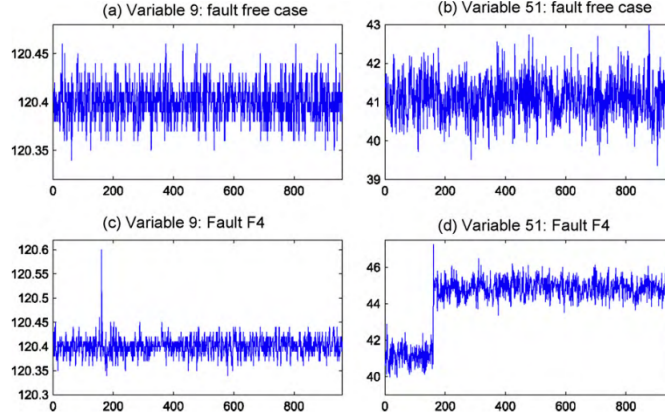


Figure 6: Tennessee Eastman : (a) Reactor temperature (X9) vs time instants under normal condition , (b) Flowrate of coolant (X51) vs time instants under normal condition ,(c) Reactor temperature (X9) vs time instants under faulty condition, and (d) Flowrate of coolant (X51) vs time instants under faulty condition. (Reproduced from [40])

The impact of violation of these assumptions will differ from one process to another.

In the next section we briefly survey information theory based approaches which avoid the first two assumptions listed above, namely of Gaussianity and

linearity of relationships.

2.2 Time delayed Mutual Information (TDMI) based method [24]

In literature, a few approaches for FDI based on information theory criteria have also been proposed. In this section, we review the work of [24] which used Information Entropy (IE) and Mutual Information (MI) for FDI. The advantage of using information theory based approaches is that they are not limited by linearity/Gaussianity assumptions. However, computing information theory metrics can be time consuming and can become infeasible if these have to be computed for large number (or large number of pairs) of variables.

The two main steps in the work of [24] are: (i) Use of information entropy to determine variables which deviate from normal operation. This includes both cause and effect variables. (ii) Computation of time-delay mutual information amongst the deviated variables to identify causal effects.

The proposed method has been implemented on three case studies [24]: (i) An Illustration Example, (ii) Tennessee Eastman process (TE), and (iii) Continuous Catalytic Reforming Facility. We briefly discuss these case studies [24]:

Problem 1: An Illustrative Example [24]

This is a simulation example. Consider a nonlinear process with two variables related as:

$$x_t = 2.8 + e_1 \quad (2a)$$

$$y_t = x_{(t-3)}(1.8 - 0.5x_{(t-3)}) + e_2 \quad (2b)$$

where e_1, e_2 are 0-mean Gaussian random variables. x, y data for 2000 time-instants (or observations) is simulated using the model with a ramp change (fault) being introduced to x between 1500^{th} to 1700^{th} instants. The simulated variables are shown in Fig.7. The first 1000 normal observation are used to compute thresholds for IE and MI. In particular, IE is computed for each variable and indicates the degree of randomness in a variable. MI is computed for a pair of variables and is a measure of association (or dependence) between the two variables. Multiple values of IE and MI corresponding to different data-windows in the first 1000 samples are constructed. Using these values, thresholds on IE and MI are obtained as the mean plus-minus three standard deviations of the resulting IE and MI values. The corresponding threshold value for IE and MI are 2.00 (lower control limit) and 0.156 (upper control limit) respectively. In real-time, the IE values are computed using a moving window approach. In Fig. 8 (a), it is observed that IE of variables x and y cross the threshold value (fall below the red colour line) at 1500^{th} instant. This indicates that a fault has occurred with variables x and y both being amongst the root cause or the affected variables. In order to find the root cause of fault event, MI between x, y is computed and is seen to exceed the threshold value (Fig. 8 (b)) from approximately 1500 instant onwards. To further determine the cause-effect

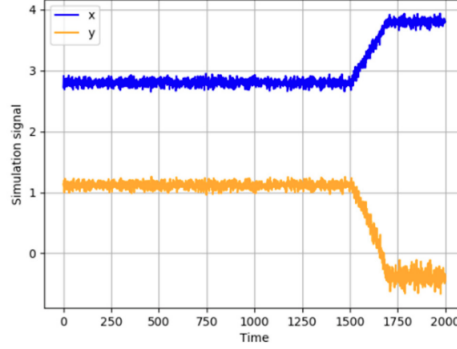


Figure 7: Illustrative Example: Simulated data. (Reproduced from [24])

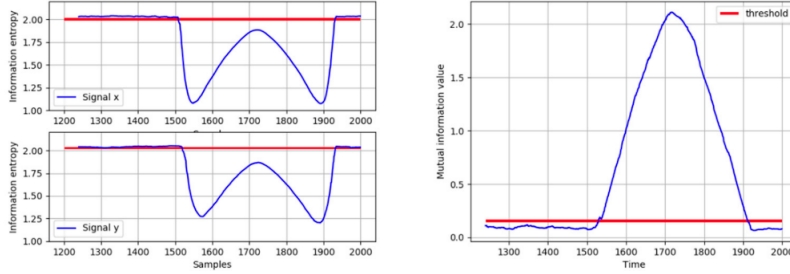


Figure 8: Simulated Example: IE and MI with threshold value. (Reproduced from [24])

behaviour between these two variables, MI value is computed for different lags. From Fig. 9, it is seen that the MI value is maximized at lag three implying that variable y is affected by values of x three instants in the past. This is consistent with the model (Eq. 2b) used to generate the data.

Problem 2: Tennessee Eastman process (TE) [24]

The proposed method is applied to TE process as shown in Fig. 4. This is also a simulation case study and the process was discussed in the previous section. The method is applied all 21 types of fault which can be simulated in the process and was able to diagnose the root cause for those cases. In particular, each fault contained 960 observations with the fault being introduced at 160th observation. Firstly, the normal observation data is used to compute IE and MI thresholds using moving windows of 100 instants. Therefore, a total of 860 IE values for individual variables, and 860 MI values for each pair of variables were computed to obtain the thresholds for these quantities. Two fault scenarios are discussed next. In the subsequent discussion- red, blue, and green represent the cause, consequent and ultimately affected node/variable.

Case 1: Fault 1: corresponds to a step change in the A/C feed ratio to the process

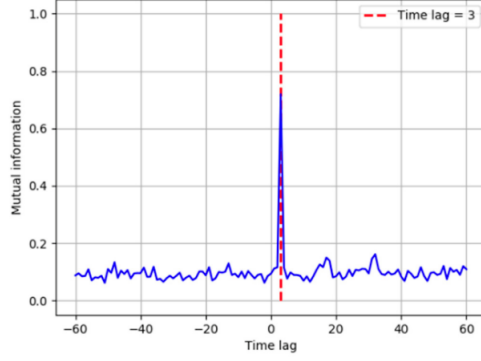


Figure 9: Illustrative Example: MI with different time lag. (Reproduced from [24])

In this case, a total of 7 variables crossed their respective IE threshold values as shown in Fig. 10a. After computing time-delayed mutual information with various lags pairwise for these 7 variables, the resulting causal digraph is presented in Fig. 10b. In this graph, the dark band on an edge indicates the directionality of the edge. In particular, it is seen that the stripper pressure (P16), compressor work (C20), and compressor recycle valve (V27) are the root causes/nodes in the digraph. While the fault corresponds to a change in A/C feed ratio, the TE simulation code made available by the creators of this problem [7] does not indicate how this is achieved. The authors of the current work being discussed [24] use first principles understanding to infer that the fault originates from the stripper pressure (P16) which is a causal node in the digraph.

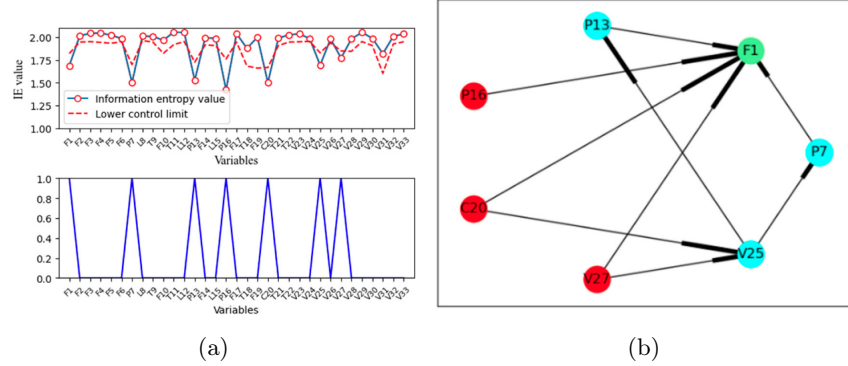


Figure 10: Tennessee Eastman: (a) IE for case 1 , and (b) Digraph for case 1. (Reproduced from [24])

Case 2: Fault 14: Sticking of Reactor cooling water valve, and Fault 17:-Disturbance in Reactor cooling water outlet temperature

Three and two variables cross IE threshold values for fault types 14 and 17 respectively as shown in Fig. 11 (a) and Fig. 11 (b) respectively. The corresponding digraph for fault type 14 and 17 is shown in Fig. 12 (a) and Fig. 12 (b) respectively. The sticking of Reactor cooling water valve affects the Reactor cooling water outlet temperature (T21) and the corresponding change of cooling water outlet temperature (T21) affects the Reactor temperature (T9). However, for fault type 17, the cooling water outlet temperature (T21) is disturbed, Hence, the controller manipulates the reactor cooling water valve to maintain the temperature constant inside the reactor. Therefore, from analysis of causality for faults 14 and 17, an interesting observation emerges namely that the causal relation can be reversed due to the controller's action.

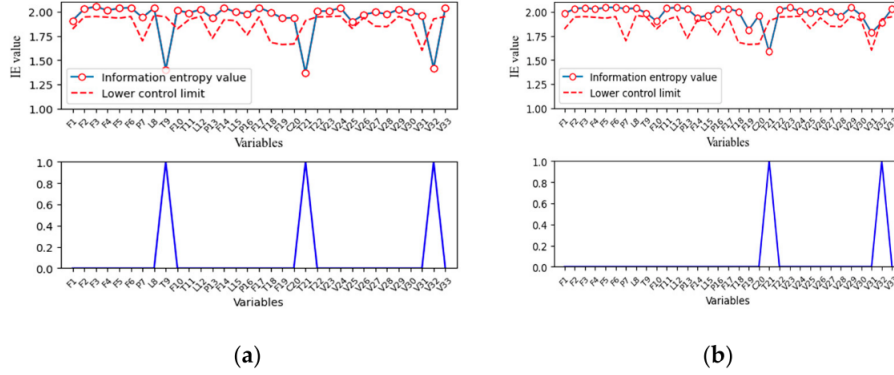


Figure 11: Tennessee Eastman: (a) IE value for fault 14 , and (b) IE value for fault 17. (Reproduced from [24])

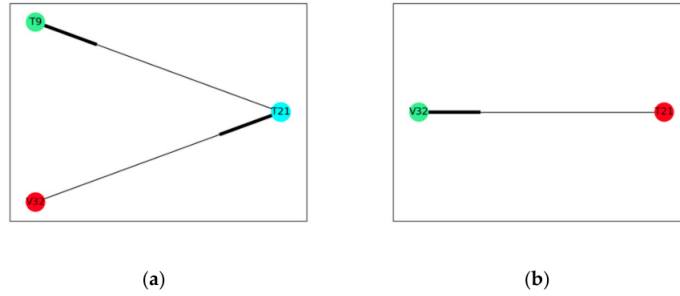


Figure 12: Tennessee Eastman:(a) Causal relation of fault 14 , and (a) Causal relation of fault 17. (Reproduced from [24])

Problem 3: Continuous Catalytic Reforming process (CCR) [24])

Unlike the previous two problems presented in [24] which were simulation studies, this problem involves analysis of industrial data. The process consists of

four reactors, four heating furnaces, and a plate heat exchanger. The heat exchanger is very importance since its performance directly affects the load of the furnace and reactor. Hence, it has significant influence on energy consumption. The pressure difference (drop) on the hot side of the exchanger is important in ensuring optimal operation and hence its unintended variation is considered to be a fault. The interest in this problem is to then identify root cause of this fault. Analysis for two different fault events has been reported in the paper [24]: (i) Varying Circulating hydrogen and (ii) Constant Circulating hydrogen.

Case (i): Varying Circulating hydrogen

The fault is detected by the information theoretic method and the identified causal relation is presented in Fig. 13a. The root cause for the Pressure drop at the hot side (PD11) is circulating hydrogen (F06) and this hypothesis can be verified from real data as given in Fig. 13b. It can be observed from 13b that increase of circulating hydrogen at the 380th sample, resulted in an increase of pressure drop at hot side at 380th sample instant.

Case (ii): Constant Circulating hydrogen

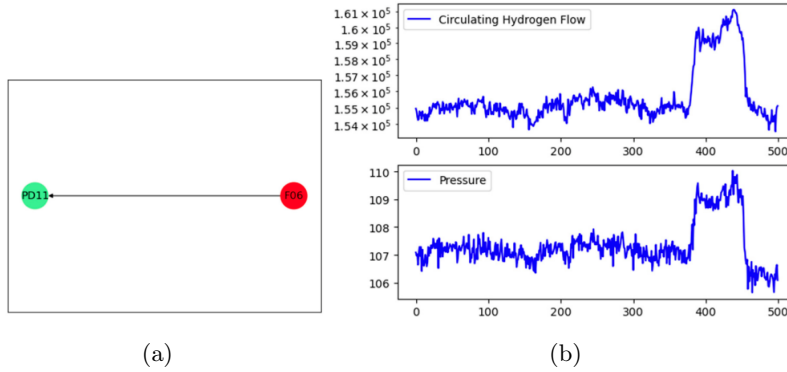


Figure 13: CCR: (a) Causal relation for case 1, and (b) Real data for case 1. (Reproduced from [24])

The fault is detected and the causal relation is given in Fig. 14a. The root cause for the Pressure drop at the hot side (PD11) is the inlet temperature at the cold side (T03). This conclusion can be verified from real data for inlet temperature and pressure drop as shown in Fig. 14b. It was observed that the inlet temperature increased as early 67th sample, and the resultant pressure drop increased at 350th sample. The outlet temperature (T04) is further influenced by increased pressure drop.

To summarize, information-theoretic approaches such as those based on Time delayed Mutual Information (TDMI), or Causation based T^2 decomposition methods for fault detection and isolation have been shown in the literature to yield promising results on several cases/illustrations including the TE challenge problem. Below we summarize the advantages and disadvantages of these

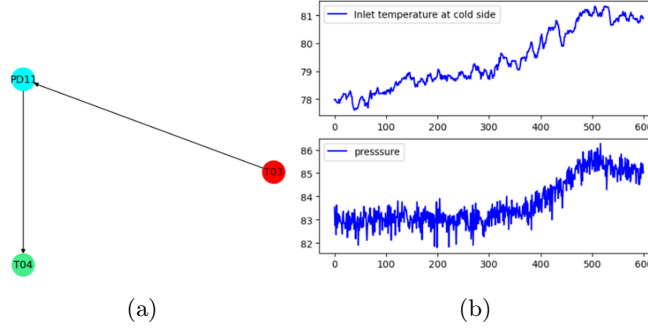


Figure 14: CCR: (a) Causal relation for case 2 , and (b) Real data for case 2. (Reproduced from [24])

approaches:

1. T^2 decomposition approach:

Advantages:

- (a) Computationally easy to apply and easy to interpret.
- (b) Explicitly incorporates the causal information captured via causal graph.

Disadvantages:

- (a) Implicitly assumes Gaussianity and linearity of relationships
- (b) Influenced by the quality of the causal graph obtained using PC algorithm (or any other approach).

2. Information theoretic approaches:

Advantages:

- (a) Do not make Gaussianity assumption and hence can be applied to data which has significantly non-Gaussian behavior.

Disadvantages:

- (a) These approaches are computationally intensive since the information-theoretic measures (entropy, mutual information, etc.) have to be computed using approaches such as kernel density estimation and numerical integration.
- (b) In the absence of predefined causal relationships, quantities such as mutual information have to be computed for large number of pairs of variables corresponding to large number of delays.
- (c) Information theory methods only look at bivariate relationships (pair-wise mutual information). Thus, given two causal relations namely

$A \rightarrow B, A \rightarrow C$, where the former ($A \rightarrow B$) effect has a larger lag, on computing pairwise mutual information, one may wrongly deduce that $B \rightarrow C$. On the other hand, the use of formal causality digraph detection tools such as PC algorithm try to avoid such spurious effects by use of conditional independence concept.

Table 1: Comprehensive Literature Survey [RCI: Root Cause Analysis, C: Classification, FD: Fault Detection, PM: Process Monitoring, IT: Information Theory]

#	Year	No. of citation	RCI/C/FD/PM	IT/Distance	Application	References
1	2021	14	RCI	MI, IE, TDMI	TE, Catalytic Reforming Facility	Ji et al.[24]
2	2017	48	RCI	T2, SPE statistics	TE	Luo et al.[32]
3	2019	33	RCI	Multiblock transfer Entropy based	TE	Zhu et al.[44]
4	2019	11	RCI	Marginal probability of faulty data	Two tank heater sys, TE	Kim et al.[28]
5	2010	122	RCI	T2	TE, Hot forming process	Verron et al.[40]
6	2024	-	C	GA	Four tank system, TE	Ardali et al.[6]
7	2022	25	RCI	T2, SPE statistics	CSTR	Ali et al.[3]
8	2016	5	C	Multivariate dyn. Kriging meta model	Three tank system	Ardakani et al.[5]
9	2018	92	PM	T2	TE	Deng et al.[20]
10	2022	27	RCI	T2, fuzzy C-means clustering KNN	TE	Elshenawy et al.[21]
11	2022	12	C	Adaptive sparrow search algo	TE	Hu et al.[23]
12	2013	2	C	KLFDA	TE	Jian et al.[26]
13	2019	90	RCI, C	Nonlinear SVM	TE	Onel et al.[35]
14	2020	12	FD	pseudo label Confidence	TE	Jia et al.[25]
15	2022	42	FD, C	-	TE	Zhang et al.[42]
16	2015	110	PM	Gaussian mixture model (GMM)-based weighted KICA (WKICA)	TE	Cai et al.[10]
17	2016	115	FD	Canonical correlation analysis ,T2	Heater, TE	Chen et al.[15]
18	2004	152	PM	Just-in-time learning PCA, T2	Simulation	Cheng et al.[16]
19	2002	154	PM	Modified distance KLJD, T2	TE	Chiang et al.[17]
20	2017	27	FD	DePCA, Bayesian inference Strategy	TE	Deng et al.[19]
21	2014	172	FD	ICA	TE	Fan et al.[22]
22	2017	298	RCI, FD	deep belief network,MI	TE	Zhang et al.[43]
23	2020	173				Md et al.[34]
24	2023	17				Ramirez et al.[36]
25	2021	67				Taqvi et al.[39]

Review Paper

3 Data Driven Modeling and Root Cause Identification Approach

Based on literature survey discussed in earlier sections, we propose the following data-driven approach for fault detection and diagnosis to be followed for the current project. This approach involves two main steps:

- Causal Network Generation: Use of Peter-Clark (PC) (or any other appropriate) algorithm to obtain a causal network or digraph, from operating data, which represents the relationships between variables.
- Fault Detection and Root Cause Analysis: Use of the derived causal digraph while applying the ' T^2 Decomposition' method to perform root cause analysis.

In summary, the process involves first constructing a causal network with the PC algorithm to identify variables (tags) contributing to the slippage event. Then, the ' T^2 Decomposition' method will be employed to determine the root cause based on the generated digraph as shown in Fig. 15. The details of PC algorithm and T^2 decomposition approach are described in subsequent subsections. It is important to note that both the T^2 metric and the PC algorithm inherently assume that the variables follow joint Gaussian distributions. However, real-world plant data are often non-Gaussian and non-stationary. As a standard practice, the data from real plants must be transformed into a Gaussian form using appropriate data transformation methods, as suggested in the literature. Therefore, a brief discussion of the data transformation method will be provided next.

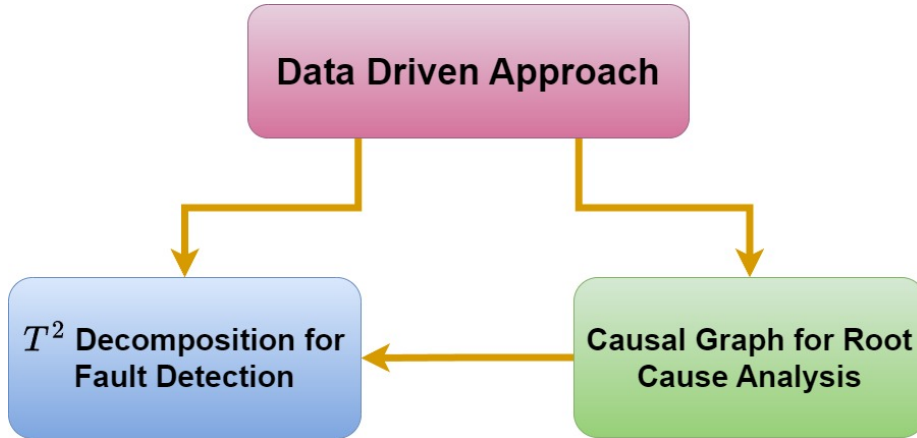


Figure 15: Data Driven Approach

3.1 Data transformation

To assess the Gaussianity of data there are several statistical tests available in literature. In this work two different tests: the Lilliefors test and the Jarque-Bera test are explored.

- The Lilliefors test is used to check if a sample comes from a normally distributed population.
- The Jarque-Bera test assesses whether the skewness and kurtosis of a sample match that of a normal distribution.

To transform the data from non-Gaussian to Gaussian several transformation methods are investigated, including the **Box-Cox Transformation**, the **Yeo-Johnson Transformation**, and the **Copula (Gaussian)** method. Among all the methods considered, we ultimately chose the **Copula (Gaussian)** method. This approach offers a significant advantage over the Box-Cox transformation, as it not only transforms multivariate non-Gaussian data into a Gaussian distribution but also preserves the dependency structure between variables. Unlike the Box-Cox transformation, which is univariate and may not capture complex, non-linear dependencies, the Copula method is specifically designed to handle multivariate relationships, making it more suitable for maintaining the joint distribution characteristics in complex datasets.

The Copula transformation is particularly effective for converting multivariate non-Gaussian data into a multivariate Gaussian distribution while preserving the dependency structure between variables. Unlike univariate methods such as the Box-Cox or Yeo-Johnson transformations, copulas handle multivariate relationships directly and can model complex, non-linear dependencies. This makes copulas especially valuable in applications where maintaining the joint distribution characteristics is crucial. Once the transformed data is obtained it is further used in PC Algorithm for digraph generation and T^2 Decomposition which is discussed next.

3.2 PC Algorithm

PC algorithm is used to extract a causal graph amongst variables given historical process data. A key difficulty in extracting relationships amongst variables in a process is the coupled nature of the process. Thus, if variable X_1 affects X_2 and X_2 in turn affects X_3 , then variable X_3 will turn out to be correlated (or dependent on) to both X_1 and X_2 . This complicates the procedure of causality extraction. The PC algorithm solves this problem by using the notion of conditional independence. In probability theory, two events A and B are conditionally independent given a third event C, if the occurrence or non-occurrence of A and the occurrence or non-occurrence of B are independent events given that event C has occurred. Conditional independence is denoted as $\{A \perp\!\!\!\perp B|C\}$.

The PC Algorithm works in three steps. It starts by constructing a complete undirected graph where every pair of nodes is connected by an edge. In the next

step, edges between pairs of variables that are conditionally independent given some subset of other variables are removed. In last step, the undirected graph is converted into a directed acyclic graph (DAG), where the directions of the edges represent causal relationships. The following example illustrates the working of PC algorithm.

Example: To illustrate PC algorithm steps [38].

Consider a problem with five variables A, B, C, D, and E. These five variables can be thought of as five tags (measurements) obtained from the plant. The objective is to extract a graph which represents the causal relation between the tags. The true graph which represents causal relation between the variables is given in Fig. 16. The steps of PC algorithm are as follows:

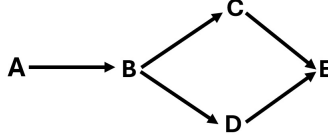


Figure 16: True graph. (Reproduced from [38])

- **Step 1:** Form a complete undirected graph for the five tags with each tag corresponding to a node in the graph. In a complete graph, all the nodes (all variables) are connected with each other. Fig. 17 shows the graph for considered example.

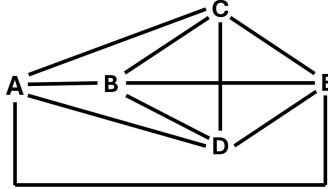


Figure 17: Complete undirected graph. (Reproduced from [38])

- **Step 2:** Eliminate edges based on conditional independence.
 1. As there is no node that is not connected to any other node, graph has no zero order independencies ($n = 0$).
 2. Now, we first examine the first order independencies ($n = 1$) where independence between carefully chosen pair of variables conditioned on a single variable is checked. From Fig. 17, it is seen that the adjacent nodes of node A are $adj(A) = \{B, C, D, E\}$. If variable A and its adjacent variable is conditionally independent, then the corresponding edge is removed from the graph. From Fig. 16 it is seen that we will have $\{A \perp\!\!\!\perp C|B\}$, $\{A \perp\!\!\!\perp D|B\}$ and $\{A \perp\!\!\!\perp E|B\}$. Hence,

the edges between A and nodes C, D, E are removed. Fig. 18 shows the resulting graph. Given B , the deleted edges $A - C, A - D, A - E$ from the graph are recorded in a set labeled as *Sepset* to be used later on. In particular, $Sepset(A, C) = B$, $Sepset(A, D) = B$ and $Sepset(A, E) = B$. We now examine the adjacent nodes of node C in

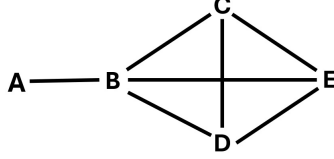


Figure 18: $n=1$, $adj(A)$. (Reproduced from [38])

Fig. 18, $adj(C) = \{B, D, E\}$. Here, $\{C \perp\!\!\!\perp D | B\}$, hence edge $C - D$ is removed and recorded as $Sepset(C, D) = B$. Fig. 19 shows the obtained graph. This procedure does not result in elimination of any other edge.

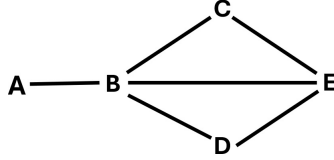


Figure 19: $n=1$, $adj(C)$. (Reproduced from [38])

3. We now examine the second-order independencies ($n = 2$). From Fig. 19, the adjacent nodes of node B are $adj(B) = \{C, D, E\}$ and $\{B \perp\!\!\!\perp E | (C, D)\}$. Hence, $B - E$ edge is removed and recorded as $Sepset(B, E) = \{C, D\}$. Fig. 20 shows the obtained graph. No further edges can be removed by this procedure.

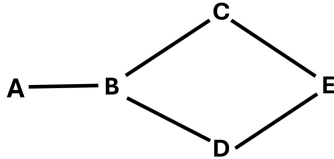


Figure 20: $n=2$, $adj(B)$. (Reproduced from [38])

4. The procedure of removing edges terminates as n cannot be increased further. Compared to the true graph (Fig. 16), it is seen that the graph structure is correctly recovered. The next step is to fix the directions of edges as far as possible.

- **Step 3:** Obtaining the direction of edges.

In this step, each set of triplets with a connecting path in Fig. 20 is considered to extract the direction of edges. In this Figure, a total of six triplets are possible. These are $\{A, B, C\}$, $\{B, C, E\}$, $\{C, E, D\}$, $\{E, D, B\}$, $\{D, B, A\}$ and $\{D, B, C\}$. The direction of the edge is obtained as per the following rule: for each triplet of vertices $\{X, Y, Z\}$ such that while the pairs $\{X, Y\}$ and $\{Y, Z\}$ are each adjacent in graph but the pair $\{X, Z\}$ are not adjacent in graph, orient $X - Y - Z$ as $X \rightarrow Y \leftarrow Z$ if and only if Y is not in $\text{Sepset}(X, Z)$. The rationale behind this rule is that since conditioning on Y was not the reason for removing edge between X and Z , the node Y cannot affect (be a cause) variables X and Z . If Y had affected X or Z , then X and Z would have been independent given Y . In Fig. 20, only triplet $\{C, E, D\}$ satisfies this condition. Hence the obtained graph is as given in Fig. 21

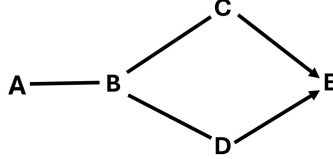


Figure 21: Obtained directed graph. (Reproduced from [38])

Compared to the the true graph, it is seen that we are able to recover the structure though direction of effect could not be recovered for all edges. This in turn implies that during usage, multiple possibilities corresponding to bidirectional edges will have to be considered.

Note: In this example, it was assumed that the independence test used to assess conditional independence will be perfect. In practice, this test will be based on available data which in-turn might introduce its own errors depending on the quality and quantity of the available data.

3.3 T^2 decomposition

An efficient fault detection and isolation tool should be able to isolate the variables implicated in the fault, in order to help the process operator to identify the root cause (the source) of the fault. As discussed in Section 2.1, one of the methods for solving this problem is based on a decomposition of the T^2 statistic. To recall, the principle of this method is to decompose the T^2 statistic in a limited number of orthogonal components which are also statistical distances. The T^2 computation is given as follows:

$$T^2 = (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \quad (3)$$

where $\mathbf{x}$ is the measurement vector assumed to be sampled from a distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. In univariate case $\mathbf{x} = x$, $\boldsymbol{\mu} = \mu$ and $\boldsymbol{\Sigma} = \sigma$, and the T^2 expression becomes:

$$T^2 = \frac{(x - \mu)^2}{\sigma^2} \quad (4)$$

In case of p variables, one way of decomposing T^2 is given as:

$$T^2 = T_1^2 + T_{2|1}^2 + T_{3|(1,2)}^2 + \dots + T_{p|(1,2,3\dots p-1)}^2 \quad (5)$$

where $T_{i|(j,\dots,k)}^2$ represents the T^2 statistic of variable X_i conditioned on variables $X_j, \dots, X_k$ i.e. the T^2 of the residual after the linear effect of $X_j, \dots, X_k$ (obtained by linear regression) has been subtracted from X_i . Note that above Equation only lists one of the possible $p!$ decomposition.

Example: The possible T^2 decomposition for a process with three variables are as follows [30]:

$$T^2 = T_1^2 + T_{2|1}^2 + T_{3|(1,2)}^2 \quad (6)$$

$$T^2 = T_1^2 + T_{3|1}^2 + T_{2|(1,3)}^2 \quad (7)$$

$$T^2 = T_2^2 + T_{1|2}^2 + T_{3|(1,2)}^2 \quad (8)$$

$$T^2 = T_2^2 + T_{3|2}^2 + T_{1|(2,3)}^2 \quad (9)$$

$$T^2 = T_3^2 + T_{1|3}^2 + T_{2|(1,3)}^2 \quad (10)$$

$$T^2 = T_3^2 + T_{2|3}^2 + T_{1|(2,3)}^2 \quad (11)$$

If the causal relationship between variables is known, then causation-based T^2 decomposition can be obtained. For example, if the causal relation between variables for this example is as in Fig. 22, then T^2 decomposition to be considered will be: $T^2 = T_1^2 + T_{2|1}^2 + T_{3|1}^2$.

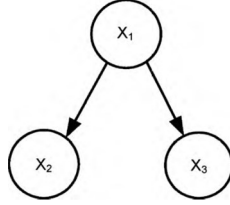


Figure 22: Example: Causal graph.(Reproduced from [30])

In general the $T_{x_1|y}^2$ value, where y contains the set of variables $\{x_2, x_3, \dots, x_n\}$, is as follows:

$$T_{x_1|y}^2 = \frac{(x_1 - \mu_{x_1|y})^2}{\boldsymbol{\Sigma}_{x_1|y}} \quad (12)$$

where, $\mu_{x_1|y} = \mu_{x_1} + \boldsymbol{\Sigma}_{x_1,y} \boldsymbol{\Sigma}_{y,y}^{-1} (y - \mu_y)$ and $\boldsymbol{\Sigma}_{x_1|y} = \boldsymbol{\Sigma}_{x_1,x_1} - \boldsymbol{\Sigma}_{x_1,y} \boldsymbol{\Sigma}_{y,y}^{-1} \boldsymbol{\Sigma}_{y,x_1}$. Here $\boldsymbol{\Sigma}$ is the covariance matrix of all variables (including x_1, y) and μ is the

corresponding overall mean vector. Further, μ_y is the mean vector of variables y , $\Sigma_{y,y}$ is the covariance matrix of variables in vector y , and $\Sigma_{x_1,y}$ is the cross-covariance of x_1 with variables in y . The “|” in notation indicates conditioning. For example, $\mu_{x_1|y}$ indicates conditional mean of x_1 given vector y .

3.4 Threshold computation

The threshold computation is used to determine whether a data point or a set of observations significantly deviates from the mean or expected value. The T^2 statistic is computed for each observation in a multivariate dataset, and it is compared to a threshold value to assess the deviation from the mean. It detects whether an observation is "out of control" or significantly different from the mean due to a faulty event. An appropriate method to compute the threshold based on Fisher distribution law (F-distribution) is given in Verron et. al. [40]. It is given as follows:

$$T_{Threshold}^2 = \frac{(m+p)(m-p)}{m(m-k-p)} F_{1-\alpha,p,m-p} \quad (13)$$

where m is the number of data points, k is the number of condition tags, p is the number of variables, and α is the significance level. Note that each term follows a Fisher distribution law.

3.5 Summary of adopted methodology

In this subsection, the overall methodology adopted for root cause analysis of slippage events is discussed. The flow chart of the adopted method is shown in Fig. 23, and its implementation includes the following two parts:

Offline step:

1. Take the historical data and perform data pre-processing by removing any missing values. Then, data clipping will be applied by setting upper and lower limits. This will give normal data.
2. Using the Copula transformation method, transform the data into Gaussian data.
3. Obtain Digraph using PC algorithm from normal operating data.
4. Determine the threshold of decomposed T^2 values for each variable.

Online step:

1. Collect the real-time data and perform the data preprocessing step.
2. Check if any of the tags has zero standard deviation of its measurement it means the measurement is faulty and the number of times it happened will be counted and stored for final statistics of root cause analysis.

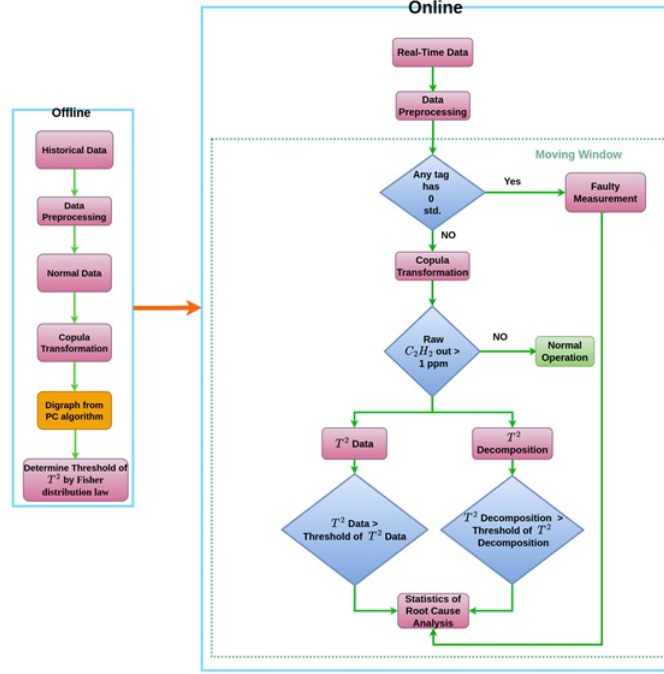


Figure 23: Flow chart of applying PC algorithm and T^2 decomposition for Root cause analysis for slippage event

3. If the measurement is not faulty, perform the copula transformation of the data to get the data with a Gaussian distribution.
4. Check if slippage occurred. If the value of $C_2H_2 out$ is less than 1 ppm, the plant is in normal operation.
5. If slippage occurs, compute the T^2 data and T^2 decomposition.
6. If the value of T^2 data is greater than its threshold it means the slippage event is captured and the count is stored.
7. If the value of T^2 decomposition for the corresponding variable is greater than its threshold it means the variable is the potential root cause of the slippage event. The count is stored in statistics.

4 Implementation of the method on illustrative Example

We now illustrate the two tools, namely causal digraph extraction using PC algorithm, and fault detection and root cause isolation through T^2 decomposition on an example. Towards this end, we assume a model relating variables, and generate data from the model by simulation. T^2 analysis and causal graph extraction is performed using the generated data. In particular, we first focus on fault detection and root cause isolation through T^2 decomposition assuming that the causal graph is correctly known. Subsequently, we discuss construction of the causal digraph with the PC algorithm.

4.1 Illustrative Example 1

To illustrate the methods for FDI problem, let us consider an illustrative example with five tags (X_1, X_2, X_3, X_4, X_5) which are related linearly as:

$$X_1 = X_2 + X_5 + \zeta_1 \quad (14)$$

$$X_2 = X_3 + X_4 + \zeta_2 \quad (15)$$

$$X_3 = X_5 + \zeta_3 \quad (16)$$

$$X_4 = X_5 + \zeta_4 \quad (17)$$

$$X_5 = \zeta_5 \quad (18)$$

where, $\zeta_1, \zeta_2, \zeta_3, \zeta_4, \zeta_5$ are normally distributed mean zero random variables representing process and measurement noise. The corresponding digraph is represented in Fig. 24.

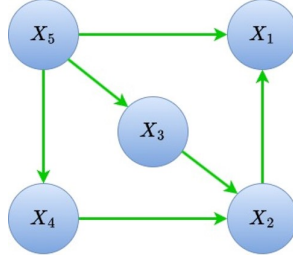


Figure 24: Digraph for five variable toy example

1. **T^2 Decomposition implementation:** For T^2 decomposition based root cause identification, we assume that the graph in Fig. 24 is known. The T^2 decomposition to be used for this illustrative example then is:

$$T^2 = T_5^2 + T_{1|(5,2)}^2 + T_{2|(3,4)}^2 + T_{3|5}^2 + T_{4|5}^2 \quad (19)$$

It is to be noted that the various terms on the right hand side in the above equation are consistent with the digraph in Fig. 24. For example, it is seen from the digraph that variable X_2 is affected (it has incoming edges) directly by variables X_3 and X_4 only. Thus, while computing T^2 corresponding to variable X_2 , it is conditioned on variables X_3 and X_4 i.e. the term $T_{2|3,4}^2$ is used instead of T_2^2 . However, variable X_5 is not affected by any other variable, and hence we see a term T_5^2 in the T^2 decomposition.

For the given illustrative example, we have simulated 1000 normal (fault-free) observations in order to find the mean vector and covariance matrix corresponding to normal data. Towards this end, random variables $\zeta_1, \zeta_2, \zeta_3, \zeta_4, \zeta_5$ are sampled 1000 times from Gaussian distributions with mean zero and standard deviations 2, 3, 4, 5, and 7 respectively. For each realization of these ζ variables, we compute $X_1, \dots, X_5$ based on Eqn. 18. Since the equations are linear, all the five tags (X_1, X_2, X_3, X_4, X_5) will have a Gaussian distribution.

To investigate the efficacy of T^2 decomposition based approach in identifying the root causes, we simulate and analyze the following two fault scenarios: (i) Fault originates from X_2 i.e. X_2 is root cause, and (ii) Fault originates from X_5 i.e. X_5 is root cause. These scenarios are discussed next.

Then we simulated 1000 abnormal (faulty) observations for two cases:

Case 1: X_2 is the root cause: For this scenario, Eqn. (15) is replaced by the following equation:

$$X_2 = X_3 + X_4 + \mathbf{20} + \zeta_2 \quad (20)$$

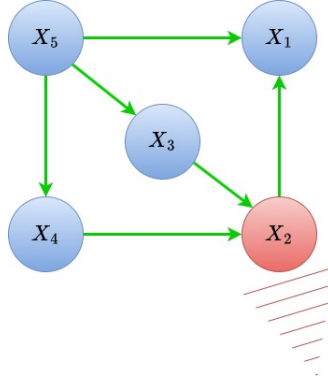


Figure 25: Illustrative Example: Digraph with fault at node X_2

The corresponding digraph is presented in Fig. 25.

Once again 1000 samples consistent with this fault behaviour are generated. The data for the five tags for both normal and fault

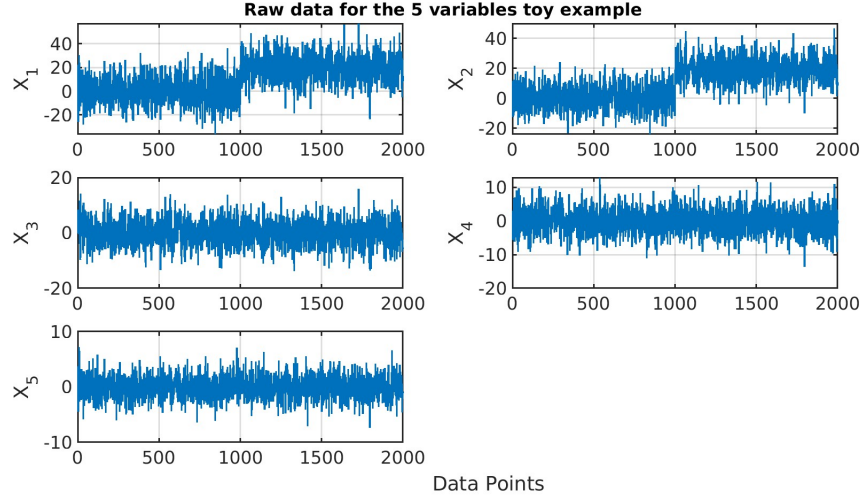


Figure 26: Plot of raw data with fault at node X_2

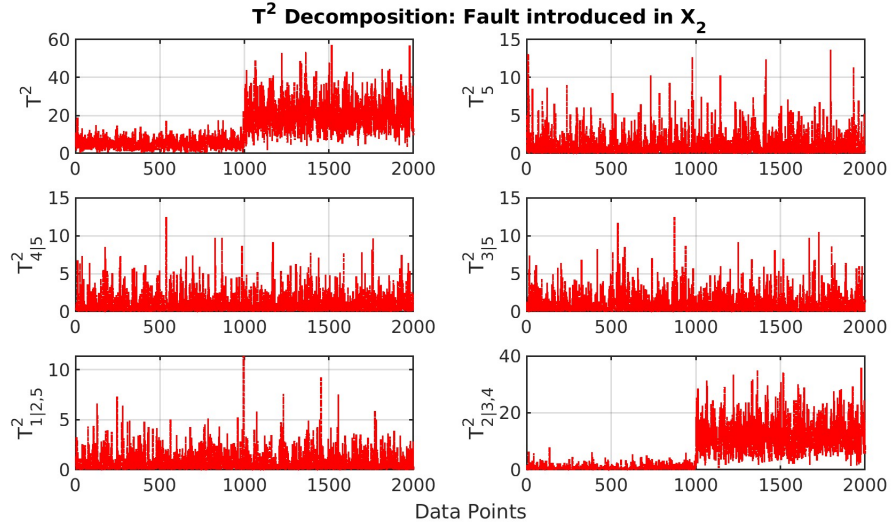


Figure 27: Plot of decomposed T^2 terms with fault at node X_2

scenario is shown in Fig. 26. In this figure, the first 1000 points correspond to normal operation while the last 1000 points correspond to fault in tag X_2 . Since X_2 affects X_1 , both variables X_1 and X_2 are deviated for the fault case. By looking at plots of this raw data, it is not possible to identify the root cause. The corresponding T^2 decomposition plots are shown in Fig. 27. Once again, the first 1000 points correspond to normal operation while the last 1000 points

correspond to fault in X_2 . The following can be observed from this Figure:

- i. Overall T^2 (top panel, first column) increases significantly once the fault occurs, thereby detecting the fault.
- ii. Amongst the individual T^2 terms only $T_{2|3,4}^2$ (bottom panel, second column) goes up after the occurrence of the fault. Note that $T_{1|2,5}^2$ (bottom panel, first column) continues to behave normally even though tag X_1 itself had higher than normal values (top panel, first column in Fig. 26).

Thus, from the plots of decomposed T^2 terms, it can be correctly concluded that fault originates from variable X_2 .

Case 2: X_5 is the root cause: For this scenario, Eqn. (15) is replaced by the following equation:

$$X_5 = \mathbf{20} + \zeta_5 \quad (21)$$

i.e. a fault of magnitude 20 units is introduced in tag X_5 . Fig. 28 shows the corresponding digraph. Once again, 1000 samples consistent with this fault behaviour are generated. Both normal and fault dat for the five tags is shown in Fig. 30 with first 1000 points corresponding to normal operation and the last 1000 points corresponding to fault originating from variable X_5 . Due to the coupled nature of the variables in the model equations, it is seen from this figure that all the variables have deviated significantly from the nominal values after the occurrence of the fault (in X_5). By looking at this raw data, it is not possible to identify the root cause. The corresponding plots of decomposed T^2 terms are shown in Figure 30. The overall T^2 shown in top panel-left column increases on occurrence of fault, thereby detecting the fault. Amongst the individual T^2 terms, only T_5^2 went up correctly indicating that the root cause is variable X_5 .

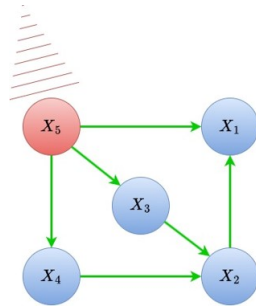


Figure 28: Illustrative Example: Digraph with fault at node X_5

The corresponding digraph is presented in Fig. 28.

2. **Generation of Causal Digraph using the PC Algorithm:** For this analysis, we generate data using the equations (Eq. 14 to 18) i.e. the

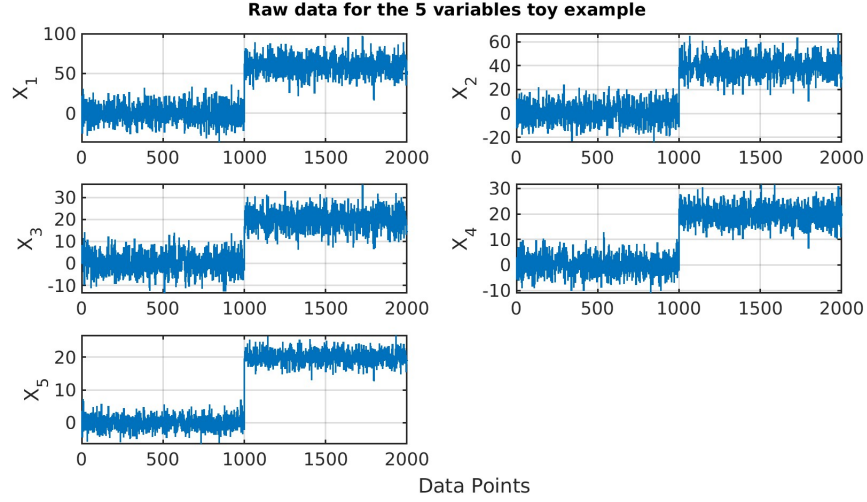


Figure 29: Plot for raw data with fault at node X_5

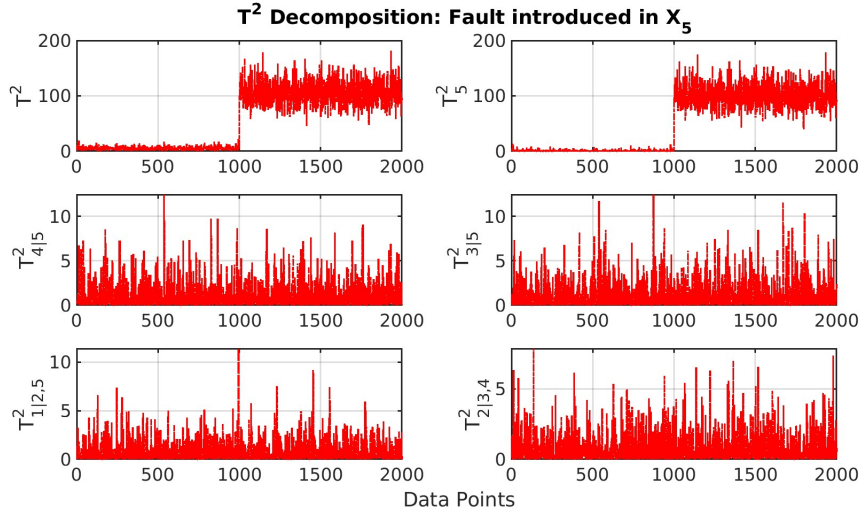


Figure 30: Plot for T^2 decomposition with fault at node X_5

model without any fault. In particular, 2000 (1st 1000 samples are with nominal data and next 1000 samples are with faulty data) samples are generated. The original causal digraph corresponding to these equations is shown in Fig. 24. By applying the PC Algorithm to the generated samples, we aim to determine how closely the original digraph can be reproduced. Towards this end, we apply two different PC Algorithm libraries: (i) one from MATLAB, and (ii) the other from the Python `causallearn` library.

Fig. 31a shows the causal digraph generated using the MATLAB library [11], while Fig. 31b displays the digraph identified using the Python `causallearn` library ([12],[13]).

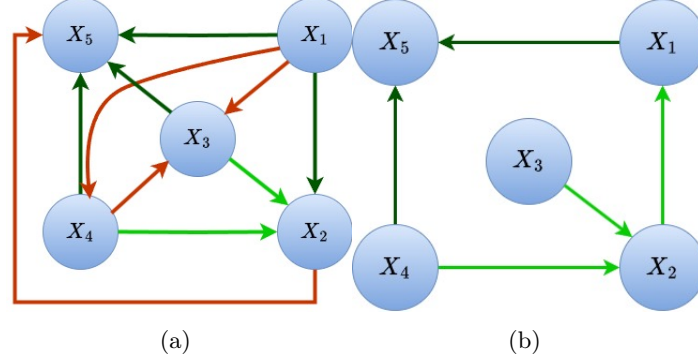


Figure 31: Digraph generation by PC algorithm (a) From MATLAB library (b) From python ‘`causallearn`’ library

It can be seen that the digraph produced by the MATLAB library (Fig. 31a) contains 10 edges, which is 3 more ((red colored edges)) than the original digraph (Fig.24). Additionally, three of these edges are in the opposite direction compared to those in the original digraph, as indicated by the dark green color.

On the other hand, the digraph produced by the Python `causallearn` library (Fig. 31b) contains no additional edges compared to original digraph (Fig.24). However, it has two edges with reversed directions (shown in dark green) and one missing edge (from X_5 to X_3).

Based on this initial preliminary analysis, it appears that the Python library performs better than the Matlab option. Apart from Matlab and Python, the R-studio also has a library named PCalg for causal graph identification. In future we will investigate performance of that library as well.

Based on this initial preliminary analysis and also the fact that Python is open-source, we propose to use the Python-based `causallearn` library for generating the causal digraph for the industrial dataset shared by Ingenero.

4.2 Illustrative Example: Two variable toy example (With nonlinear data)

In the previous section, generation of process digraph, and effectiveness of T^2 decomposition approach based on process digraph to diagnose root cause of fault was investigated. However, T^2 metric implicitly assumes that the variables have joint Gaussian densities which is equivalent to saying that the underlying relationships amongst variables are linear (Section 2.1). In this section, we investigate the effectiveness of the T^2 method for non-linear data. It is to be noted that in literature the T^2 decomposition method has been implemented on ‘Tennessee Eastman dynamic simulation process which is highly non-linear in nature. For this process, the T^2 decomposition method was able to perform root cause analysis for the fault diagnosis quite well [41]. In this section, we investigate the performance by generating simulated data for a toy example.

As a first step, we will assess the extent of non-linearity present in the plant data. We will then apply a similar level of non-linearity to a toy example to determine whether the T^2 decomposition remains effective under the non-linear relationships.

1. Plant nonlinearity analysis:

Based on first principle knowledge, we have identified several variables from the plant data that could be contributing to the slippage event. These variables are: Feed CO, H_2 -in, C_2H_4 -in, T -in, C_2H_2 -in, and the H_2/C_2H_2 inlet molar ratio. We now examine the nature of the relationship between C_2H_2 -out with each of these six variables. In particular, we obtain plot of C_2H_2 -out with these variables based on the minute-wise data from time-period 23rd January 2024, 21:18 to 25th January 2024, 23:17, total 3000 data points in Reactor-A. The 7th cycle comprises a total of 1,45,380 data points, from November 21st, 2023, 00:00 to February 29th, 2024, 22:59. Within this range, the data exhibits highly non-linear behavior. Therefore, to investigate whether the data is slightly non-linear, we consider small patches of the dataset for a more detailed examination of its behavior as shown in Fig. 32. It should be noted that this is an approximate analysis since all the six variables were varying simultaneously, and hence exact relationship between C_2H_2 -out and a specific variable cannot be directly teased out just by observing its behaviour vis-a-vis the specific variable. We aim to replicate the behavior of the plant data using an illustrative example.

2. Nonlinear toy example:

Next, we generate data for two variables using a toy example that exhibits a level of non-linearity similar to that observed in the plant data. In particular, X_1 and X_2 are the two variables that are related to each other as shown in Fig. 33 and as governed by Eqns. 22-23.

$$X_1 = \zeta_1 \quad (22)$$

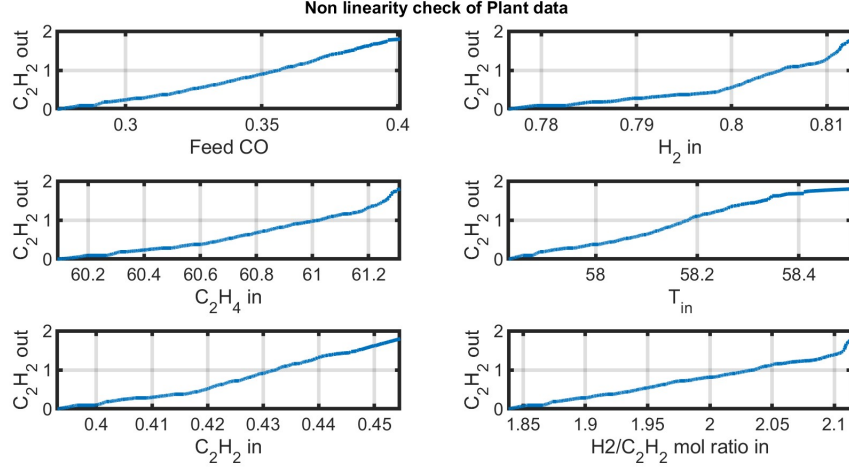


Figure 32: Non-linearity check for plant data

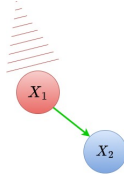


Figure 33: Digraph for two variable example

$$X_2 = X_1 + 0.1\sin(X_1) + \zeta_2 \quad (23)$$

In these equations, ζ_1 , ζ_2 are zero mean uniform random variables with variance 0.0833 unit. Using Equations Eq.22 and Eq.23, we have plotted the relationship between X_1 and X_2 for 1,000 data points, as shown in Fig. 34. This relationship exhibits a non-linearity similar to that observed in the plant data (Fig. 32).

We now evaluate whether the T^2 decomposition method effectively detects faults in the presence of this type of non-linearity. Now, after the first 1,000 normal operation samples, a fault is introduced in X_1 with a magnitude of 0.5 units, affecting the subsequent 1,000 samples as shown in Fig. 35. The corresponding data generation equations are:

$$X_1 = 0.5 + \zeta_1 \quad (24)$$

$$X_2 = X_1 + 0.1\sin(X_1) + \zeta_2 \quad (25)$$

We observe that after introducing a fault of 0.5 units, X_1 shifts by 0.5 units, and a similar shift is also seen in X_2 as shown in Fig. 35. Thus,

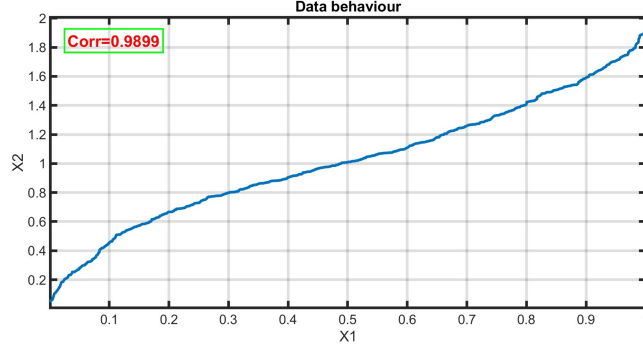


Figure 34: Non-linearity check for 2 variable toy example

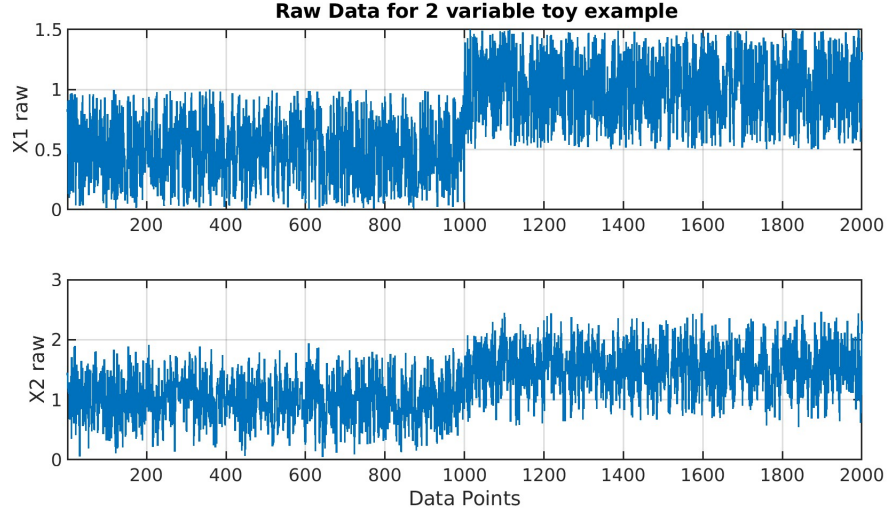


Figure 35: Raw data plot for two variable toy problem

based on visual analysis of raw observations, the root cause cannot be diagnosed. Figures 36 and 37 show the corresponding behaviour of the overall and decomposed T^2 values.

Observation: As seen in Figure 36, the T^2_{Data} increases after 1000 samples, indicating the occurrence of a fault. Figure 37 further reveals that X_1 is the root cause of the fault. This conclusion is derived based on the observation that $T^2_{X_1}$ increases significantly after 1000 samples, while $T^2_{X_2|X_1}$ remains unchanged i.e. consistent with the normal data. From this, it is seen that T^2 decomposition method performs effectively with mildly non-linear data. It should however be noted that if the extent of non-linearity is increased, then the method may not cleanly diagnose the

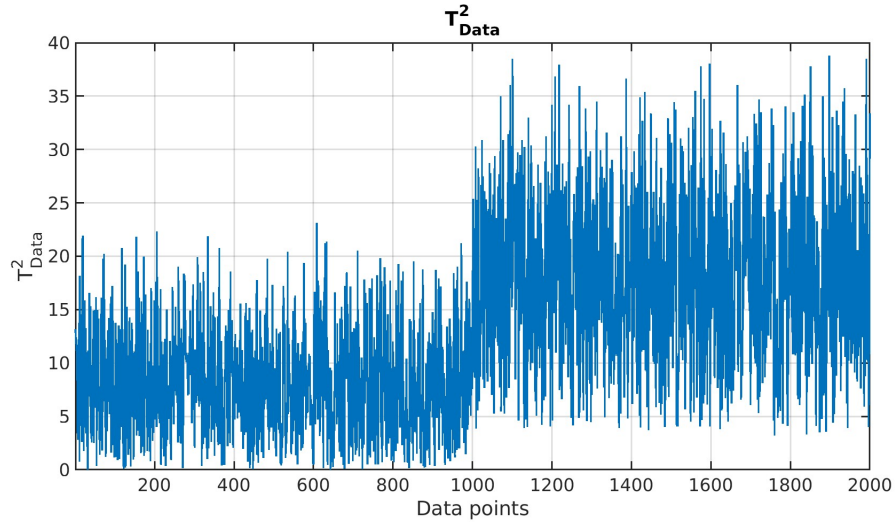


Figure 36: T^2 plot for two variable toy problem

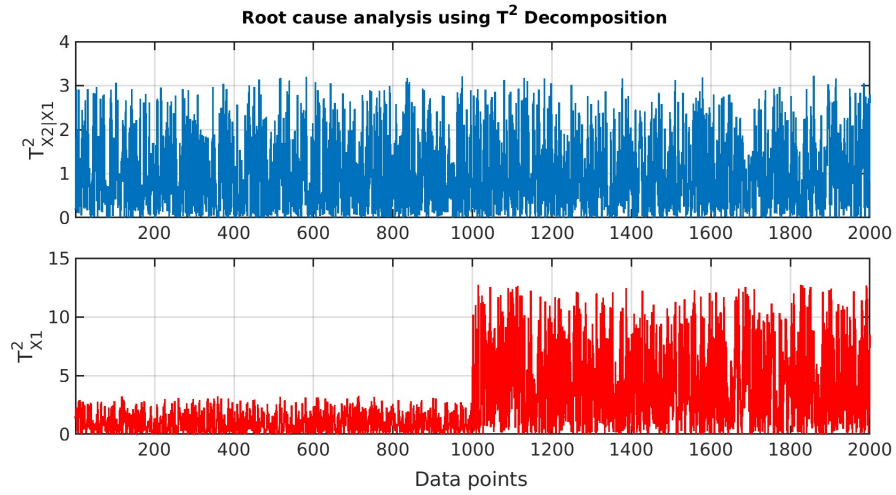


Figure 37: T^2 decomposition plot for two variable toy problem

root cause. To demonstrate this, a new simulation has been performed with increased non-linearity, and the results are presented in Figure 38. For this simulation, the same set of Equations (Eqn. 22 - Eq. 25) have been used. However, in Equation 23 and in Equation 25, the value of 0.1 was replaced with 1, and the simulation is performed again. The corresponding results from this updated simulation are shown in Figure 38.

Figure 38d shows that with increased non-linearity, the T^2 decomposition is unable to clearly identify the fault. This is evident as $T^2_{X_2|X_1}$ also demonstrates intermittent higher values after the 1000th sample point, as compared to its behaviour under normal operation (1st to 1000th sample). This will lead to X_2 also being identified as a root cause apart from X_1 at some time instants.

It should be noted that in practise, while plants are nonlinear, in a small window of operation the plant behaviour can be considered to be approximately linear.

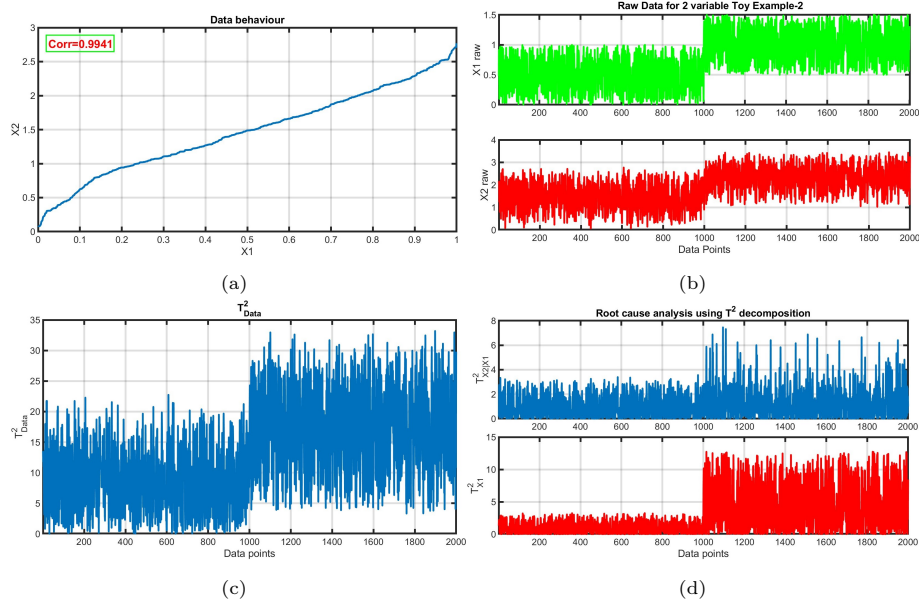


Figure 38: Fig.(a) Non-linearity check for 2 variable toy example, (b) Raw data plot for two variable toy problem, (c) T^2 plot for two variable toy problem, (d) T^2 decomposition plot for two variable toy problem

Based on the analysis in this section, we now apply the T^2 decomposition method to the plant data and present the results in the following section.

4.3 Copula transformation steps illustrated with toy example:

1. Plant Data – Non-Gaussian Data:

Begin with the raw data, which is non-Gaussian in nature. Fig. 39 shows a non-Gaussian dataset generated with 1,000 samples, each having an amplitude ranging from 0 to 8 units. This is randomly generated data that has non-gaussian behavior. The respective histogram plot is shown in Fig. 40.1.

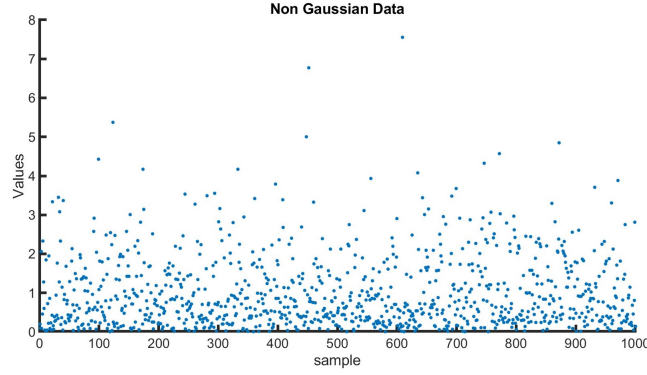


Figure 39: Illustrative Example: Toy example data

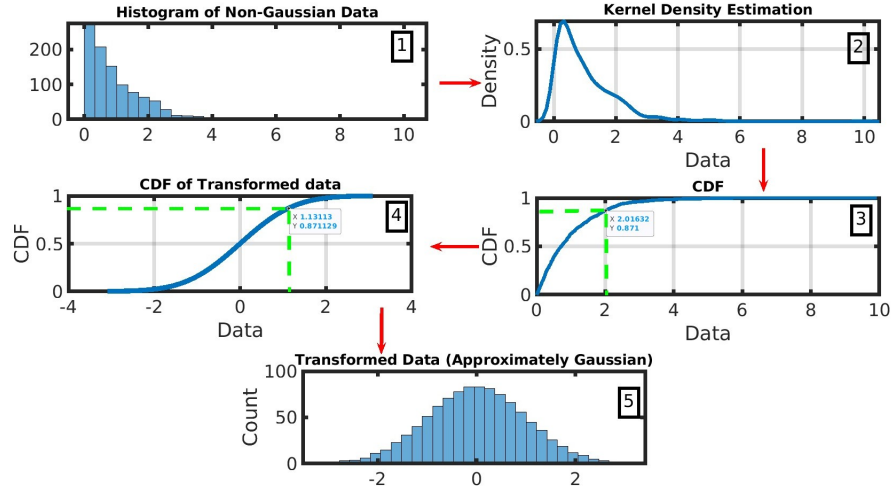


Figure 40: Illustrative Example: Copula transformation stepwise

2. Compute Kernel Density:

Estimate the probability density function of the non-Gaussian data using kernel density estimation (KDE) to capture the data distribution more accurately. Fig. 40.2 shows the kernel density estimate of the non-Gaussian data

3. Compute CDF:

Derive the cumulative distribution function (CDF) from the kernel density estimate (KDE) which is uniform in nature. Fig. 40.3 shows the CDF of KDE.

4. Transform Data Using Inverse Gaussian CDF:

Apply the inverse Gaussian CDF to the uniform data obtained from the

previous step. This step converts the data into a Gaussian distribution while maintaining the original dependency structure. Finally, Fig. 40.4 displays the transformed CDF, and the histogram plot of the transformed data is shown in Fig. 40.5.

5 Acetylene Reactor Data Preprocessing

In this section, we briefly discuss the data preprocessing steps applied to the Acetylene reactors data provided by Ingenero.

- The data set provided by Ingenero has 285 tags including Process Indicator (PI) and Inferred where both Reactor A and Reactor B data have been merged. As a first step we have separated the data of both reactors cycle-wise. The resulting cycle-wise information is given in Table 2.

Table 2: Date wise data separation

	Reactor A	Reactor B
Cycle-1	14/2/2017 8AM to 31/7/2017 3AM	1/8/2016 6PM to 14/2/2017 9AM
Cycle-2	18/1/2018 12AM to 3/7/2018 1AM	30/7/2017 9PM to 18/1/2018 1AM
Cycle-3	19/12/2018 8PM to 14/6/2019 11AM	3/7/2018 12AM to 20/12/2018 2AM
Cycle-4	31/12/2019 5AM to 17/6/2020 1AM	14/6/2019 5AM to 31/12/2019 7PM
Cycle-5	12/1/2021 4AM to 6/9/2021 8AM	16/6/2020 6PM to 12/1/2021 7AM
Cycle-6	19/7/2012 8AM to 4/8/2023 12PM	6/9/2021 7AM to 19/7/2022 9AM
Cycle-7	21/11/2023 11AM to 5/6/2024 11PM	4/1/2023 11AM to 21/11/2023 12PM

- After separating the data for both reactors (Reactor A and Reactor B) and discussing with the Ingenero team, we narrowed down the total of 285 tags to 14 tags.
- From the remaining tags, based on first principles knowledge, the potential tags that may be the cause of the slippage event have been identified and are listed in Table 3. The table also list the upper and lower bound of various tags as provided.

Table 3: Tags most likely responsible for slippage events

Tag Description	PI/Inferred	Unit	L.Bound	U.Bound
Reactor Common Inlet C2H2	PI	%	0	0.8
Reactor Common Inlet H2	PI	%	0	1.7
Reactor Common Inlet Temperature	PI	°C	50	75
Rct A out T	Inferred	°C	70	100
Rct total feed	Inferred	TPH	130	200
Feed CO	Inferred	PPM	-	-
H2 C2H2 mol ratio	Inferred		1	2.5
Rct A top bed median T	Inferred	°C	-	-
Rct A mid bed median T	Inferred	°C	-	-
Rct A bott bed median T	Inferred	°C	-	-
RCT H2 flow	Inferred	KG/HR	-	-
dT	Inferred	°C	-	-
Rct H2 Out Ppm	PI	PPM	-1	1500
Rct Acetylene Out	PI	PPM	0	10

- If any row in the dataset contains a missing value, the entire row has been removed. This means the missing value data minute is removed from all other stages to make it consistent. The total missing data is 0.2% for minutewis dataset provided by the Ingenero team.

- Using the 14 tags outlined in Table 3, a preliminary analysis of slippage and selectivity has been carried out for the dataset associated with reactor A. For brevity, figures are not included in this report. This analysis is significant as it provides the only preliminary opportunity to visually examine how both slippage and selectivity are affected by changes in the values of the other 13 tags. By observing this variation, we can assess the impact of increasing the value of one specific tag on the overall slippage and selectivity. Essentially, the analysis allows us to understand how fluctuations in one tag's value influence the slippage and selectivity, offering insights into the relationships and dependencies among the tags.

6 Final implementation of the method on Acetylene Reactor dataset

This section presents and discusses the final implementation of the T^2 decomposition approach applied to Acetylene reactor data. In the preliminary implementation of the method on the Acetylene Reactor dataset, the digraph is generated from the first principle model, and based on heuristic knowledge only six variables are considered. The T^2 decomposition and root cause analysis was done with hourly as well as minute-wise data for considered six variables. The detail discussion of preliminary work is moved to Appendix A. In the final implementation of the adopted method, the minute-wise dataset of Reactor A is analyzed and the listed 14 tags in the previous section are considered. The minute-wise data shared with us is of reactor A, 7th cycle from 21st November 2023, 00:00 to 29th February 2024, 22:59, i.e. a total of 1,45,380 data points. After removing the missing value, 1,45,021 data points remain, containing 4,601 slippage events. The implementation begins by constructing a causal digraph, where 14 candidate tags most likely responsible for slippage events were selected based on heuristic knowledge. These tags were identified as the most probable contributors to slippage events. The resulting causal digraph is then utilized for T^2 decomposition and root cause analysis to investigate slippage occurrences. Details of the 14 variables considered in this study is already discussed in the previous section. The Acetylene reactor dataset provided was non-Gaussian and non-stationary, therefore as discussed in section 3.1 firstly data is transformed using the copula transformation method which is discussed next. In the preliminary stage of developing the methodology for this work, the digraph generation, T^2 decomposition, and root cause analysis were done with the raw dataset which was not transformed. The discussion of which is moved to Appendix B. As the PC algorithm and T^2 decomposition assume data to be Gaussian, it is observed that the performance of the method improved using transformed data. Hence, the transformation of data is adopted as an integral part of the methodology.

6.1 Copula transformation of data

The copula transformation is done for each of the 14 variables. Figs. 41 to 42 show the histogram plot of each variable for its Non-Gaussian and transformed data. It can be observed from the plot that, for most of the variables the copula transformation can transform the data into Gaussian data. This transformed data is further used for digraph generation and T^2 decomposition.

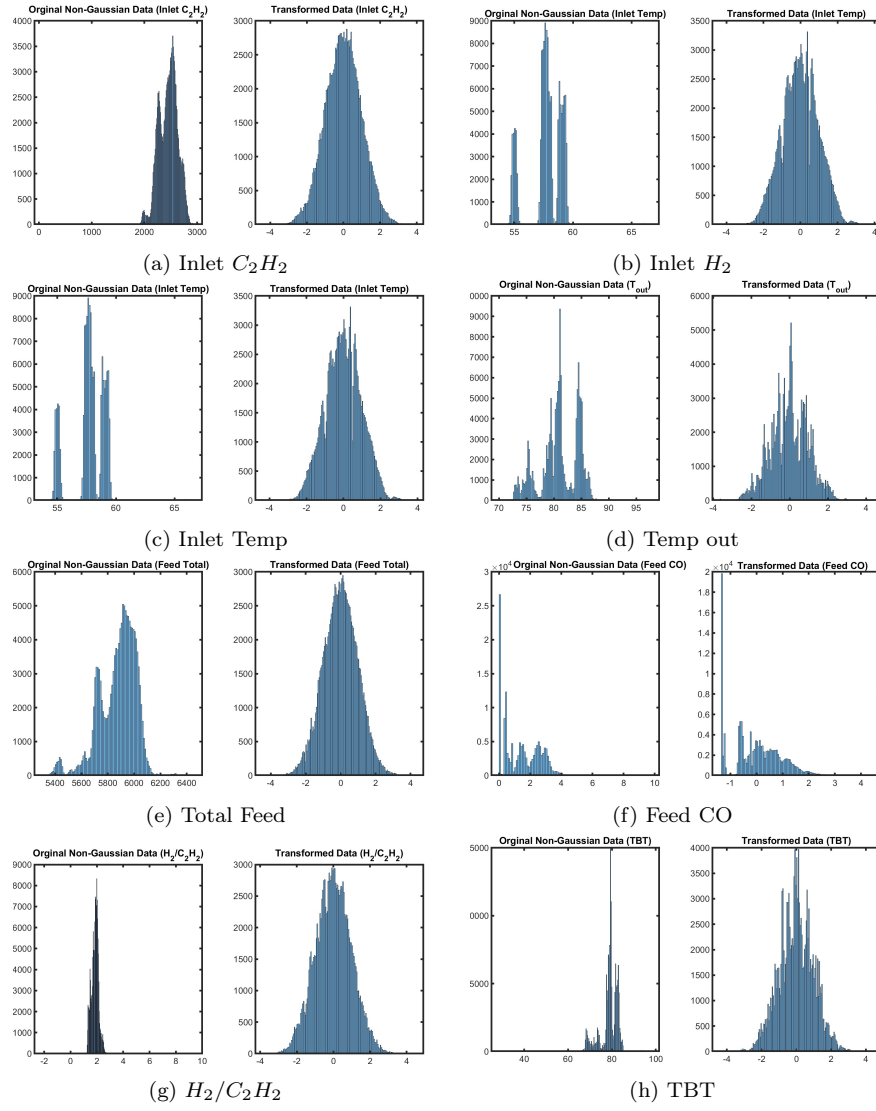


Figure 41: 14 variables without and with copula transformed

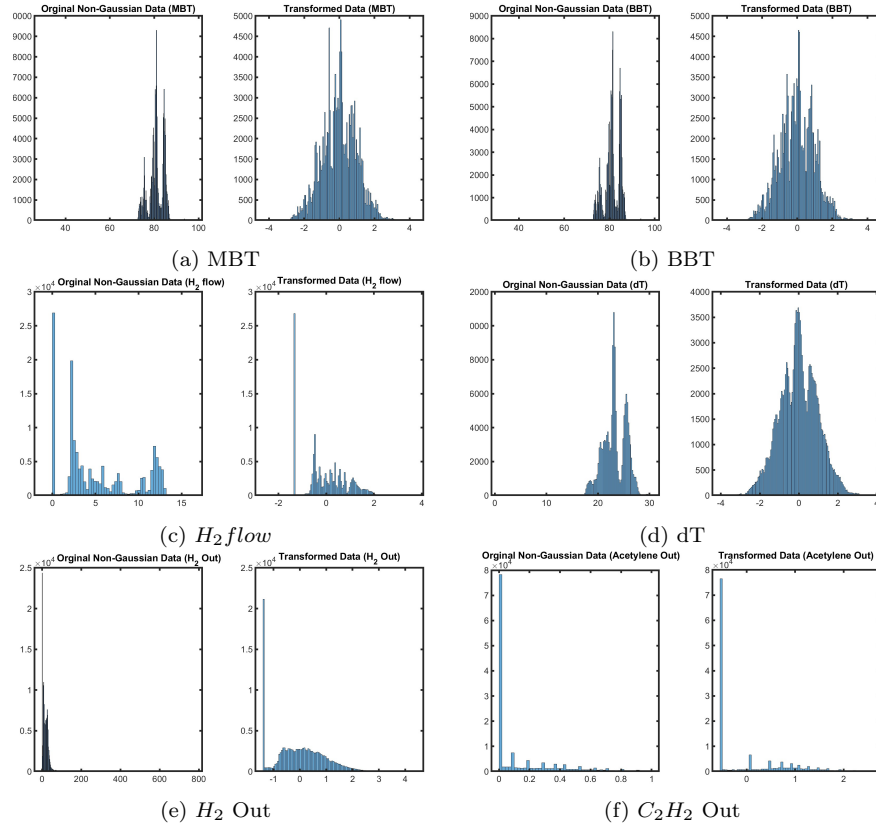


Figure 42: 14 variables without and with copula transformed

6.2 Digraph generation from transformed data

The Gaussian data obtained after the transformation for the considered 14 variables is further used to develop a digraph using the PC algorithm. Fig. 43 shows the obtained digraph. To improve the readability of the graph each node is assigned with new name i.e. X_i , where i takes the numbers 1 to 14 which corresponds to the original names of tags. It is important here to note that while developing the digraph a specific approach is adopted to remove the absurd edges which does not support the first principle knowledge. In this approach, among the input variables, intermediate variables, and output variables, a restriction is imposed for the absence of edges between few nodes. The table 4 shows the list of variables grouped into input, intermediate, and output variables. Fig 44 illustrates the restriction imposed. The nodes between which there are no edges are depicted among the input variables, intermediate variables, and output variables there is a restriction imposed. Whereas a two-sided arrow shows there is no restriction and PC algorithm will decide the presence of edge and its direction.

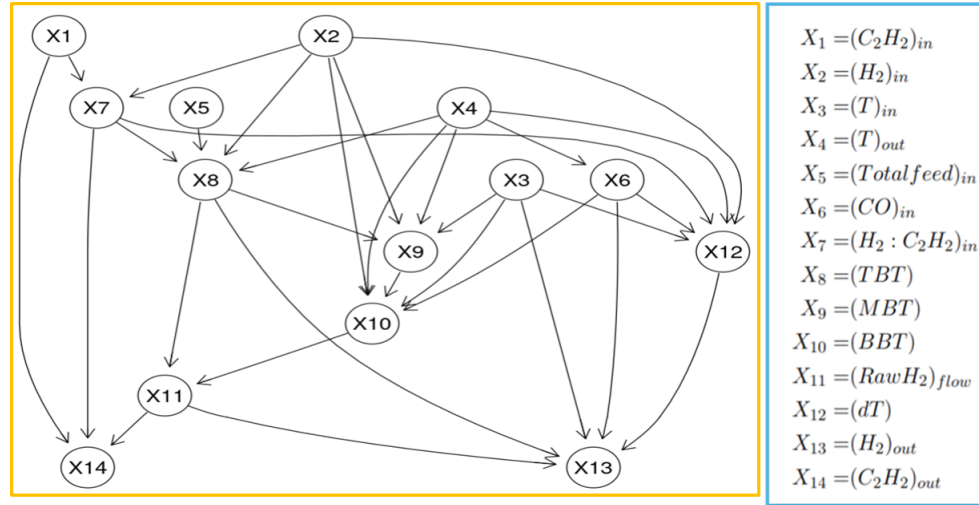


Figure 43: Digraph using transformed data with considered 14 variables

Table 4: List of variables grouped into input, intermediate and output variables

Input variables	Intermediate variables	Output variables
C_2H_2 In (kmol/hr)	Top bed temperature ($^{\circ}C$)	H_2 Out (ppm)
H_2 In (kmol/hr)	Mid bed temperature ($^{\circ}C$)	C_2H_2 out (ppm)
T In ($^{\circ}C$)	Bottom bed temperature ($^{\circ}C$)	Reactor temperature out ($^{\circ}C$)
H_2/C_2H_2 mol ratio	dT ($^{\circ}C$)	
Reactor feed total (kmol/hr)		
CO In (kmol/hr)		
Raw H_2 flow		

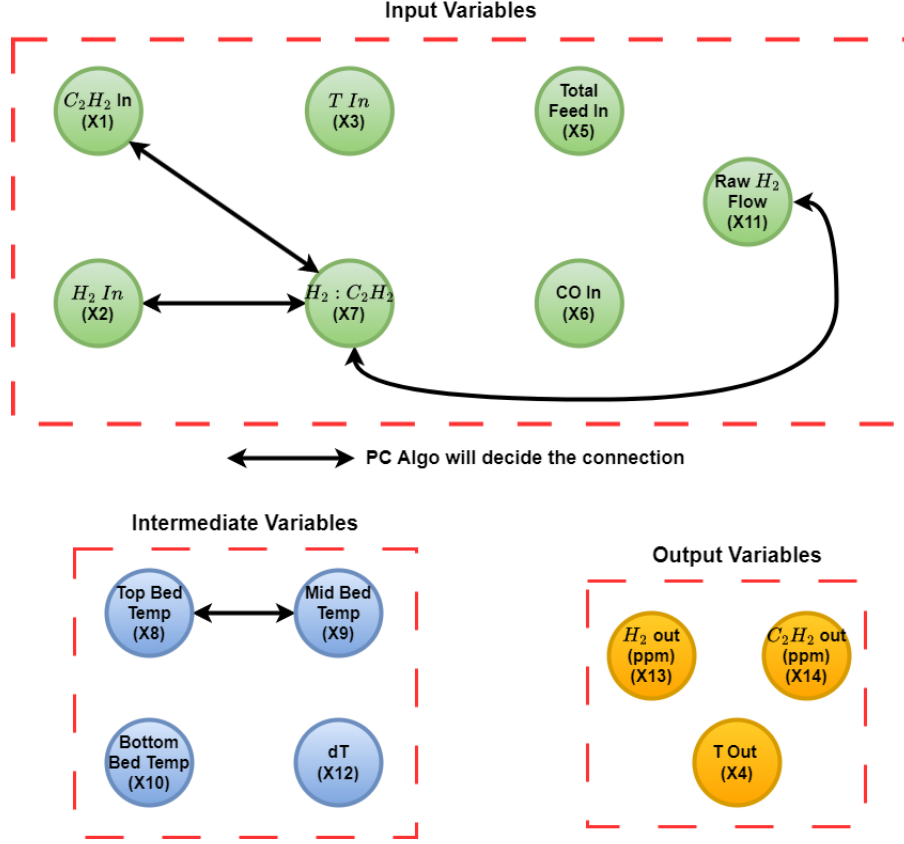


Figure 44: Approach for digraph generation

6.3 Window wise T^2 computation with transformed data

Based on obtained causal network or digraph discussed in the previous subsection, the decomposition of T^2 statistics is done as explained in section 3.3 with transformed data. The T^2 computation requires the first and second moments of the data. Therefore, instead of processing the entire dataset in a batch, a sliding window approach is adopted, with each window having a length of $50n$, where n is the number of tags considered. This parameter n is a tuning parameter, chosen heuristically. The window slides by one sample at each sampling instant. The expression for the decomposition of T^2 statistics based on the

obtained digraph and considered 14 variables is given as follows:

$$\begin{aligned}
T^2 = & T_{X_1}^2 + T_{X_2}^2 + T_{X_3}^2 + T_{X_4}^2 + T_{X_5}^2 + T_{X_6|X_4}^2 + T_{X_7|(X_1,X_2)}^2 + \\
& T_{X_8|(X_2,X_4,X_5,X_7)}^2 + T_{X_9|(X_2,X_3,X_4,X_8)}^2 + T_{X_{10}|(X_2,X_3,X_4,X_6,X_9)}^2 + \\
& T_{X_{11}|(X_8,X_{10})}^2 + T_{X_{12}|(X_2,X_3,X_4,X_6,X_7)}^2 + \\
& T_{X_{13}|(X_3,X_6,X_8,X_{11},X_{12})}^2 + T_{X_{14}|(X_1,X_7,X_{11})}^2
\end{aligned}$$

The threshold value for each of the 14 decomposed T^2 of variables is computed using the expression discussed in subsection 3.4. The ultimate objective of this project is to find the root cause of the slippage event. The thorough investigation of the provided acetylene reactor dataset shows that when slippage occurs it is sustained for a considerable amount of time. Therefore, multiple slippage events can be seen as an episode and the whole dataset can be decomposed into multiple episodes for root cause analysis. In the next subsection, the root cause analysis of one of the episodes is discussed.

6.4 Episode-wise root cause analysis of slippage event with transformed data

The episode considered for root cause analysis is from 950th instant to 1050th instant of time in minutes as shown in Fig 45. The first slippage event that occurred in this episode is at 1026th (12 Dec 2023, 19:03) instant, and the last slippage event at 1041th (12 sec 2023, 19:17). To investigate the root cause

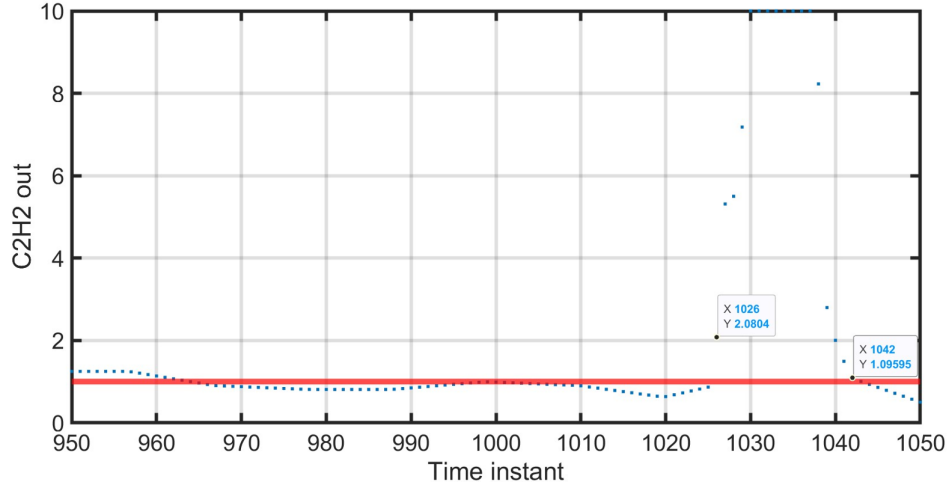


Figure 45: C_2H_2 out data, red line indicates 1 ppm

of the slippage event a specific approach is adopted in which we analyze T^2 decomposition values of each variable 30 to 35 instants back in time from when

the first slippage event of the episode occurred. Fig. 46 illustrates the above-mentioned approach. This approach is useful for online monitoring, providing the operator with an early warning several minutes before a slippage event occurs, allowing them to take corrective action promptly.

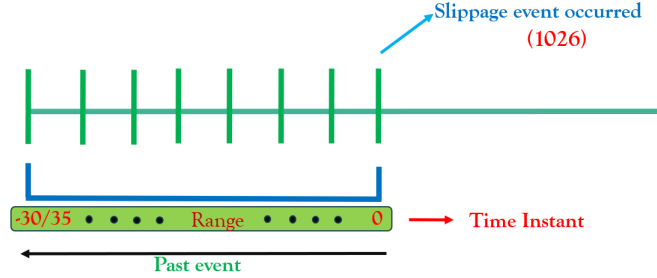


Figure 46: Episode-wise approach for root cause analysis of slippage event

Fig. 47 to Fig. 51 shows plots of Raw data (left) and T^2 decomposition (Right) vs time instant for transformed data. The blue continuous line in the right side plots are values of T^2 at every instant and instances with pink dots depict slippage events. The red constant line is for the computed threshold value. Inspecting T^2 decomposition values of each variable in Fig. 47 to Fig. 51, it can be observed that before the first slippage occurred at 1026th instant in this episode, T^2 decomposition values of C_2H_2 In, T In, $Total$ Feed, CO In and Raw H_2 has crossed their corresponding threshold whereas T^2 decomposition values of other variables are within the threshold limit. This indicates that the above variables are potential root causes of slippage events in this episode. The raw data for the above variables show significant deviation from their mean value when the T^2 decomposition values are above the threshold which confirms that the above variables are the potential culprit. Fig. 52 summarises the above analysis for this episode.

Fig. 53 shows an indication of abnormal behavior of an individual variable when a slippage event occurs and T^2 decomposition value has crossed its threshold in this episode. This plot can be very useful for real-time monitoring of the variables. Fig. 54 shows counts of the number of times T^2 decomposition of all individual tags and T^2 data cross their corresponding threshold when slippage occurs and there is no faulty measurement. Based on the above analysis some statistics on the T^2 decomposition method are presented next.

6.5 Episode-wise root cause analysis statistics with transformed data

For assessing the efficacy of T^2 decomposition methodology adopted in this work few metrics are computed with the help of a specific logical approach. Fig. 55 shows that when there is no faulty measurement and the slippage is occurred, if value of T^2 data crosses its threshold, it is said that slippage event is detected.

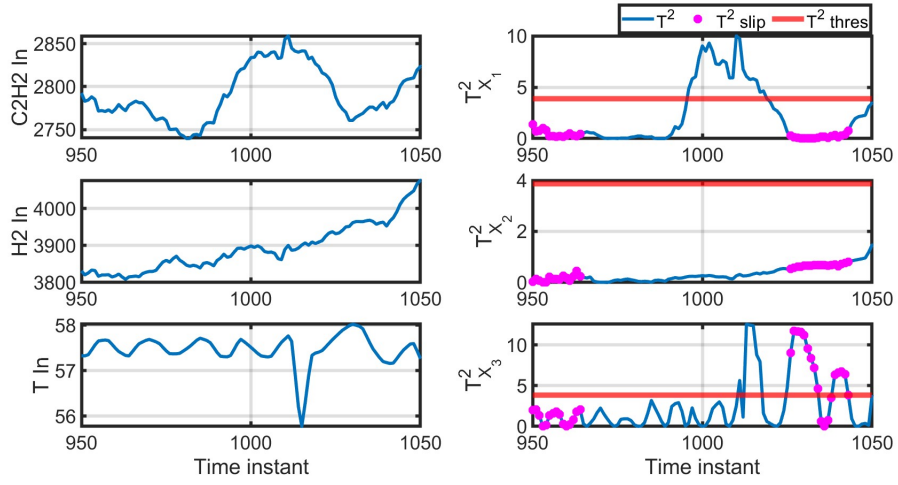


Figure 47: Plot of Raw data (left) and T^2 decomposition (Right) vs time instant for transformed data: (Top) C_2H_2 in, (Mid) H_2 in, and (Bottom) T in

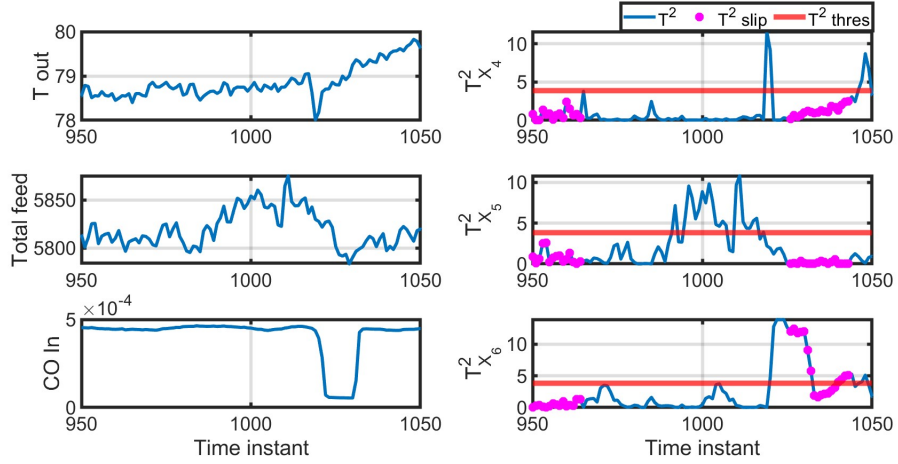


Figure 48: Plot of Raw data (left) and T^2 decomposition (Right) vs time instant for transformed data: (Top) T out, (Mid) Total feed, and (Bottom) CO in

Fig. 56 shows that when there is no faulty measurement, slippage has occurred, and T^2 of C_2H_2 out has not crossed its threshold, only T^2 of other variables has crossed the threshold, in that case, the event is said to be fully explained because variable which crossed the threshold is the root cause of slippage. In this case, if none of the other variables has crossed the threshold, it is said that the event is uncaptured. In the same Fig. 56, when there is no faulty measurement, slippage has occurred, and T^2 of C_2H_2 out has crossed its threshold and T^2 of any other

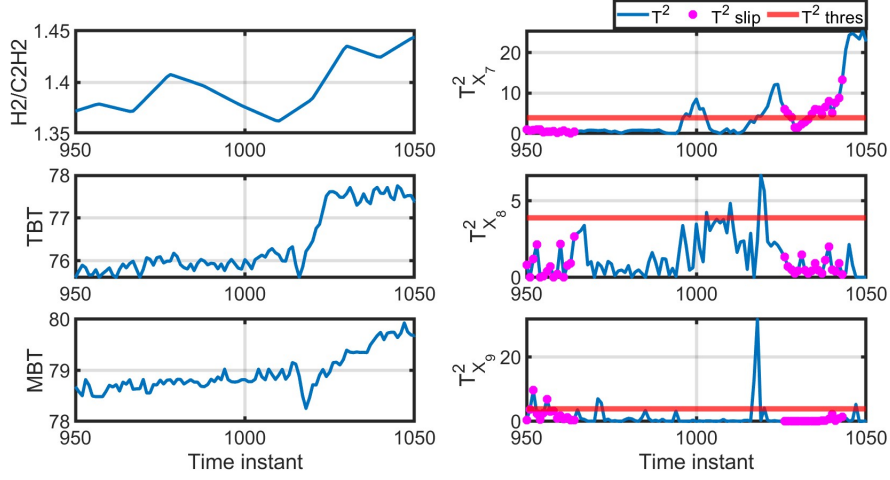


Figure 49: Plot of Raw data (left) and T^2 decomposition (Right) vs time instant for transformed data: (Top) $H_2 : C_2H_2$ in, (Mid) TBT, and (Bottom) MBT

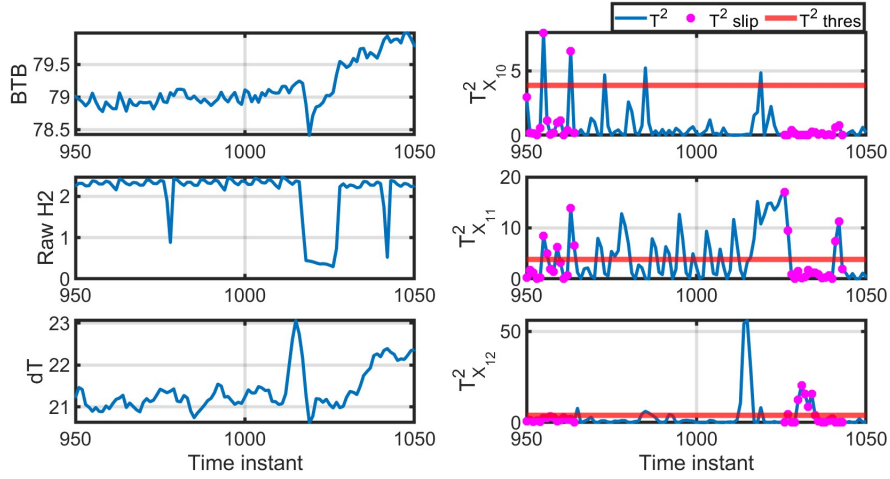


Figure 50: Plot of Raw data (left) and T^2 decomposition (Right) vs time instant for transformed data: (Top) BTB, (Mid) Raw H_2 flow, and (Bottom) dT

variables has crossed the threshold, in that case, the event is said to be partially explained because along with the variable which crosses the threshold, there is another possible variable which is responsible for slippage event but those are not modeled. In this case, if none of the other variables has crossed the threshold, it is said that the event is unexplained. Based on the above logical decisions the counts for different metrics are tabulated for considered episode 950th instant to 1050th instant in Table 5. The table shows that out of 32

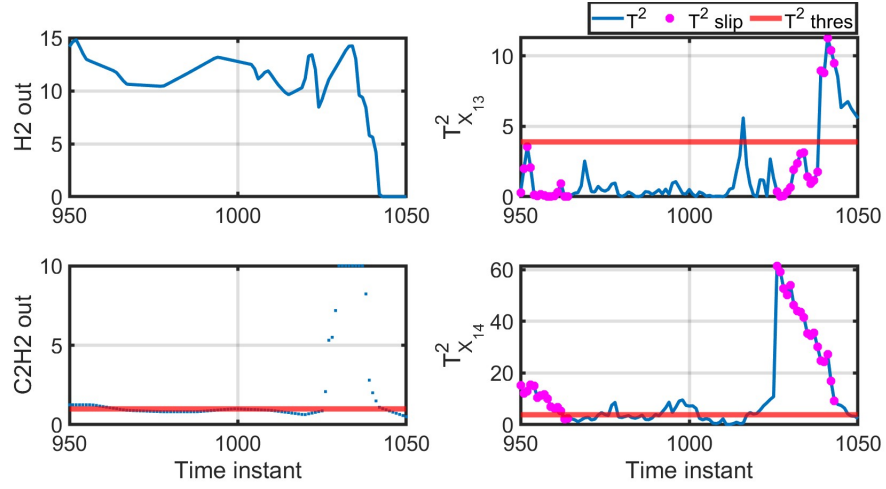


Figure 51: Plot of Raw data (left) and T^2 decomposition (Right) vs time instant for transformed data: (Top) H_2 out, and (Bottom) C_2H_2 out

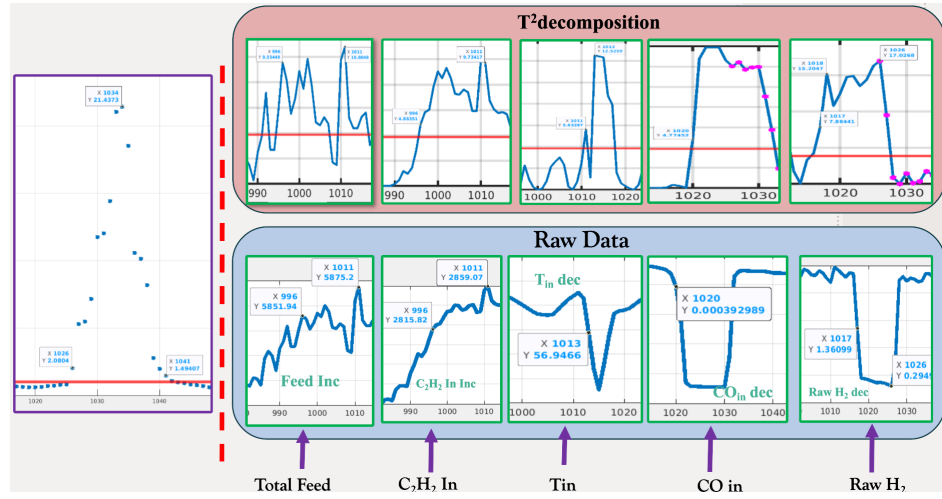


Figure 52: The potential root cause variables for slippage event in the considered episode with transformed data

captured slippage events, 6.25% of time the method can explain the root cause and 68.75% of time can partially explain the root cause. Meaning, most of the time the root cause of slippage event is explained in this episode.

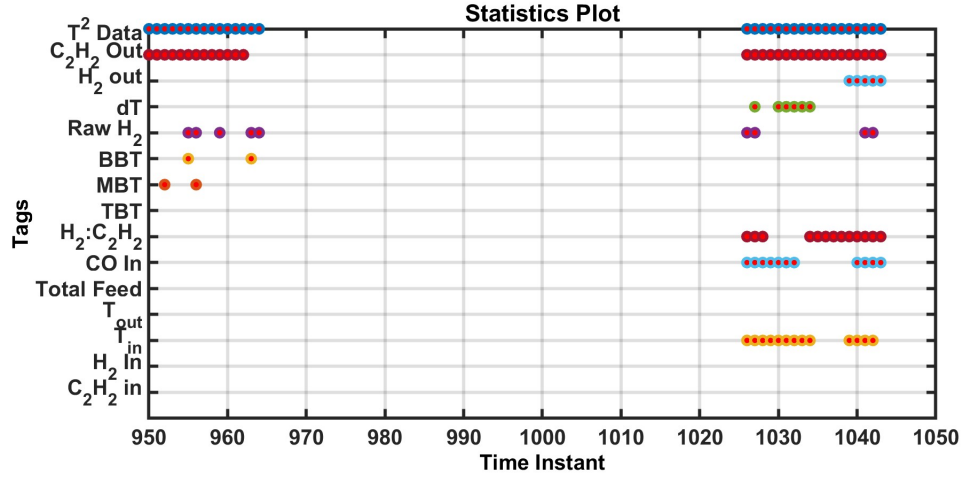


Figure 53: Indicator plot: Real-time indication of abnormal behavior of an individual variable when a slippage event occurs of transformed data.

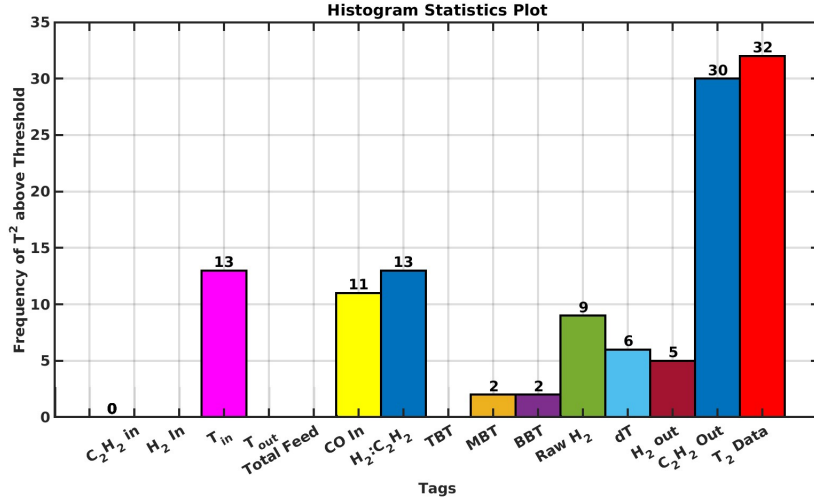


Figure 54: Histogram plot: The number of times T^2 decomposition of all individual tags and T^2 data cross their corresponding threshold for the specified range of transformed data.

6.6 Comparison of root cause analysis statistics for transformed and without transformed complete dataset

The minute-wise data provided to us pertains to Reactor A, covering the 7th cycle from 00:00 on November 21, 2023, to 22:59 on February 29, 2024, totaling

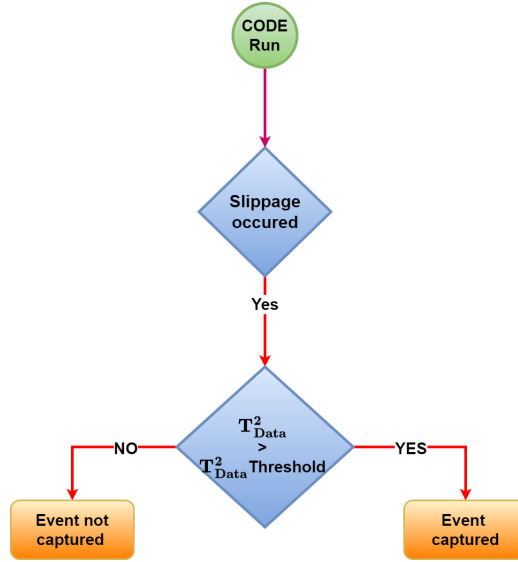


Figure 55: Flow chart showing logic for counting whether slippage event is captured by T^2 data

Table 5: Root cause analysis statistics of episode 950th instant to 1050th instant

Parameter	Statistics
Total number of data points	101
Total Slippage event	32
Total slippage event captured by T2 Data	32 (100%)
Faulty measurements	0%
Partially explained	22 (68.75 %)
Unexplained	7 (25%)
Fully explained	2 (6.25 %)
Uncaptured case	0 (0 %)

144,500 data points. After removing the missing values, 145,021 data points remain, which include 4,601 slippage events. Out of these many data points, there are 777 faulty measurements. Hence, the effective number of slippage events is 3824. The histogram of T^2 data in Fig. 86 shows that 3824 times the slippage event is captured. Other histograms show that the number of times T^2 decomposition of all individual tags has crossed their corresponding threshold for the complete data.

In the initial stage of developing the methodology for this work, the digraph generation, T^2 decomposition, and root cause analysis were performed using the raw, untransformed dataset, refer to Appendix B. Table 6 shows the comparison of root cause analysis statistics without and with transformed data. From the table, it can be observed that, with transformed data, the total number of partially explained events decreased and fully explained events increased. Also,

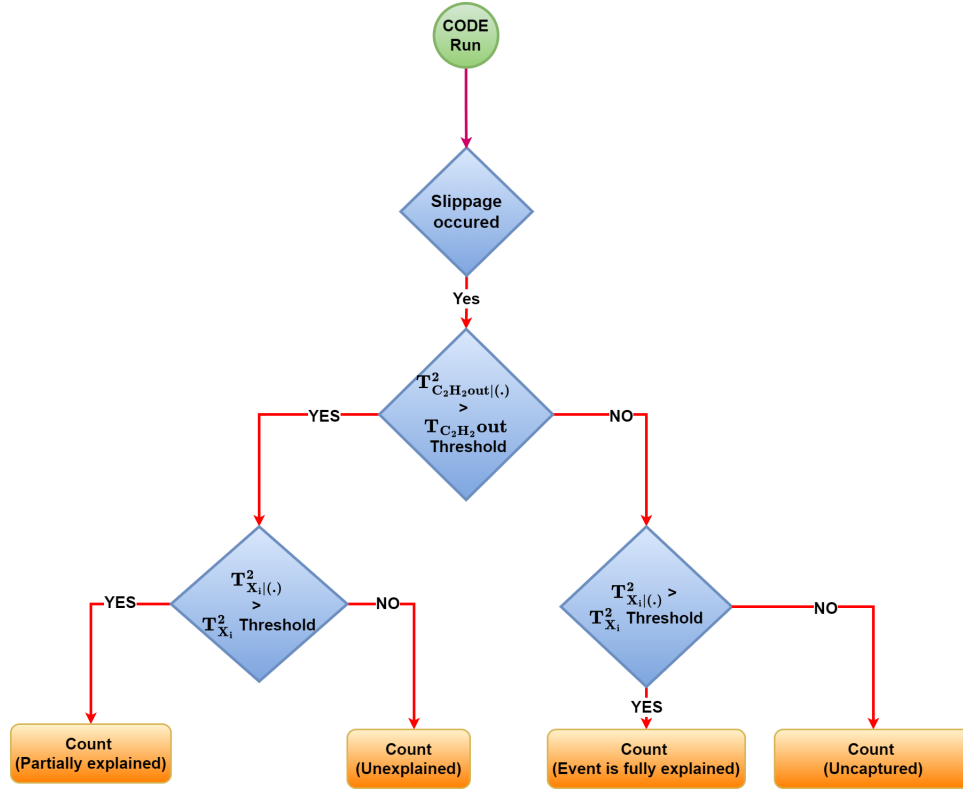


Figure 56: Flow chart showing logic for counting partially explained, unexplained, fully explained and uncaptured events of slippage

the unexplained events decreased with transformed data. Therefore it can be mentioned that the performance of the method improved using transformed data.

Table 6: Comparison of root cause analysis statistics for without and with transformed data

Statistics	Without transform	With transform
Total number of data points	1,45,021	1,45,021
Total Slippage event	4601	4601
Faulty measurement	777	777
Effective total Slippage event	3824	3824
Total slippage event captured by T2 Data	3824 (100%)	3824 (100%)
Partially explained	2583 (67.54%)	1627(42.56%)
Unexplained	638 (16.68%)	302 (7.89%)
Fully explained	497 (13%)	1433 (37.47%)
Uncaptured case	106 (2.77 %)	462 (12.08%)

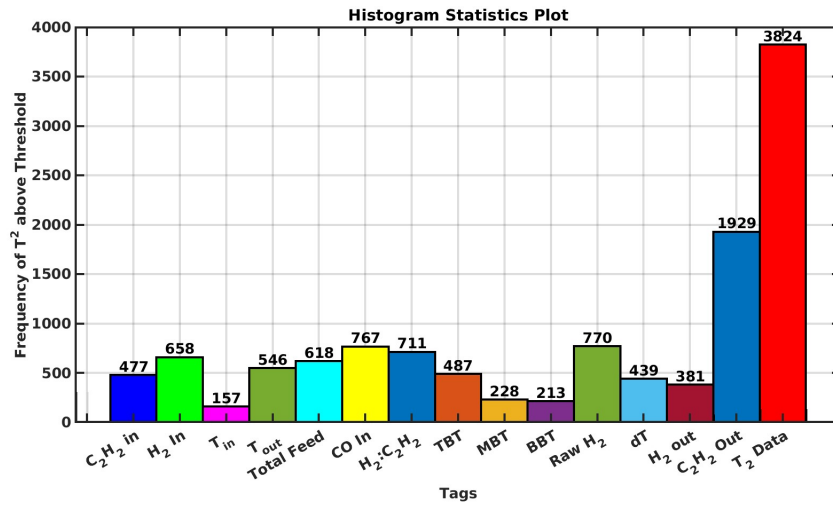


Figure 57: Histogram plot: The number of times T^2 decomposition of all individual tags and T^2 data cross their corresponding threshold for complete data range of transformed data.

7 Conclusion

Based on the literature survey, the ' T^2 Decomposition' method has been identified as a promising approach for analyzing the root cause of the slippage event. Since ' T^2 Decomposition' is a fully data-driven method, it requires a causal digraph to decide variables to be considered in T^2 decomposition. To generate the necessary causal digraph between the variables, the 'PC Algorithm' has been chosen from literature. It is important to note that both the T^2 metric and the PC algorithm assume that the variables follow joint Gaussian distributions. However, real-world plant data are often non-Gaussian and non-stationary. Therefore, as a standard practice, data from real plants should be transformed into a Gaussian form using suitable transformation methods, as recommended in the literature. Among all the methods investigated, we ultimately selected the Copula transformation method. This approach provides a notable advantage over others, as it not only converts multivariate non-Gaussian data into a Gaussian distribution but also maintains the dependency structure between variables. It is observed from table 6, that the performance of the method improved using transformed data.

The ultimate objective of this project was to identify the root cause of the slippage event. A thorough investigation of the provided acetylene reactor dataset revealed that when slippage occurred, it was sustained for a considerable amount of time. As a result, multiple slippage events were considered as an episode, and the entire dataset was decomposed into several episodes for root cause analysis. The episode considered for root cause analysis spanned from the 950th to the 1050th time instant, measured in minutes. To investigate the root cause of the slippage event, a specific approach was adopted in which the T^2 decomposition values of each variable were analyzed 30 to 35 time instants prior to the occurrence of the first slippage event in the episode. This approach is valuable for real-time monitoring, as it provides the operator with an early warning several minutes before a slippage event occurs, enabling timely corrective action. By inspecting the T^2 decomposition values of each variable in Fig. 47 to Fig. 51, it was observed that before the first slippage event at the 1026th instant in this episode, the T^2 decomposition values of C2H2 In, T In, Total Feed, CO In, and Raw H2 had exceeded their respective thresholds, while the T^2 values of other variables remained within the threshold limits. This indicated that the aforementioned variables were potential root causes of the slippage events in this episode. A similar analysis can be done for other episodes to find the root cause of the slippage event. To evaluate the effectiveness of the T^2 decomposition methodology used in this work, several statistics were calculated using a specific logical approach that classify the slippage events into fully explained, partially explained, unexplained, and uncaptured cases.

Figure 58 summarises the proposed offline and online steps of applying the PC algorithm and T^2 decomposition for Root cause analysis for slippage event. Note that the moving window size can be tuned in the offline phase based on performance on training/validation data if that is possible. Otherwise, it can be tuned to maximize the number of windows where data has Gaussian

behavior. Further, if additional tags not considered earlier, need to be added to the analysis at any point in time, then the offline procedure to generate a causal digraph has to be reapplied. Similarly, the digraph can be regenerated if a significant amount of additional normal data is available by appending this additional data to earlier available normal data.

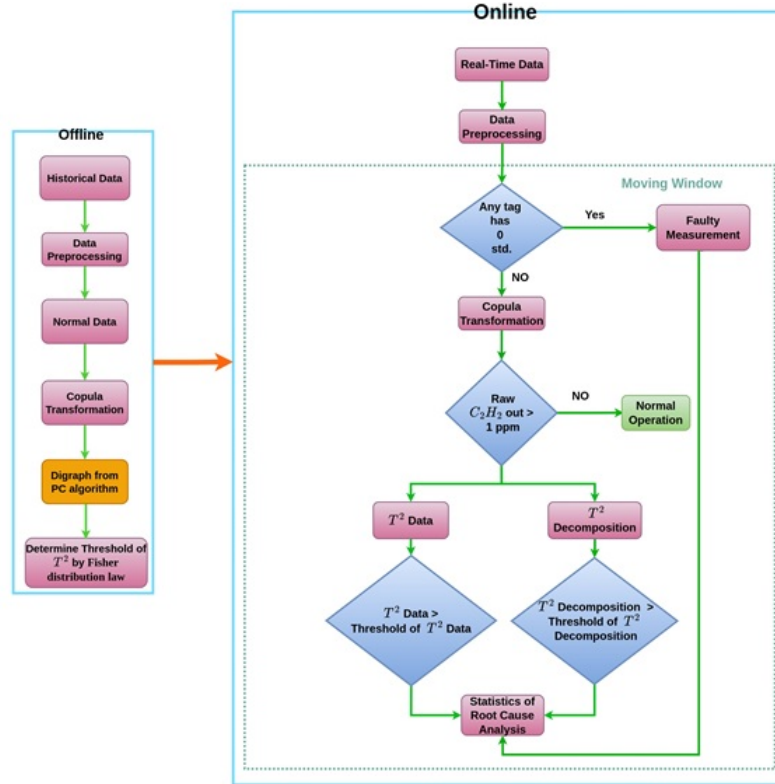


Figure 58: Flow chart of applying PC algorithm and T^2 decomposition for Root cause analysis for slippage event

A Appendix: Preliminary implementation of the method on the Acetylene Reactor dataset

In this section, some sample results obtained after a preliminary implementation of T^2 decomposition approach to Acetylene reactor data are presented and discussed. Specifically for this implementation, Reactor A has been considered. To implement the T^2 decomposition on this dataset, the first step involves obtaining a causal-based digraph. At this stage, we have established a knowledge-based causal relationship between the slippage event and five other tags instead of a data-driven approach. This is discussed next

A.1 Digraph generation from first principles model

Assuming pseudo-steady-state, the ODEs for component mass balance along the length of the reactor as considered in First Principle modeling report :

$$u_g \frac{dC_i}{dz} = \rho_s \sum_j r_{(j,i)} \quad (26)$$

where, $i = [C_{H_2}, C_{C_2H_2}, C_{C_2H_4}, C_{C_2H_6}, C_{C_4H_8}, C_{C_6H_{12}}, C_{CO}]$ and $r_{(j,i)}$ represents the rate of j^{th} reaction for i^{th} species. These rate expressions for various reactions are listed next:

- Acetylene hydrogenation reaction:

$$r_1 = \frac{a_A k_1 p_{C_2H_2} p_{H_2}}{(1 + K_{C_2H_2}^A p_{C_2H_2} + K_{CO}^A p_{CO})(1 + K_{H_2}^A p_{H_2})} \quad (27)$$

- Ethylene hydrogenation reaction:

$$r_2 = \frac{a_E k_2 p_{C_2H_2} p_{H_2}}{(1 + K_{C_2H_2}^E p_{H_2} + K_{C_2H_4}^E p_{C_2H_4} + K_{H_2}^E p_{H_2} + K_{CO}^A p_{CO})^3} \quad (28)$$

- Butene formation reaction:

$$r_3 = \frac{a_A k_3 p_{C_2H_2} p_{H_2}}{(1 + K_{C_2H_2}^A p_{C_2H_2} + K_{CO}^A p_{CO})(1 + K_{H_2}^A p_{H_2})} \quad (29)$$

- Hexene formation reaction:

$$r_4 = \frac{a_A k_4 p_{C_2H_2} p_{H_2}}{(1 + K_{C_2H_2}^A p_{C_2H_2} + K_{CO}^A p_{CO})(1 + K_{H_2}^A p_{H_2})} \quad (30)$$

The relation between partial pressure (p_i) and concentration (C_i) of component i is given by (Dalton's law) :

$$p_i = \frac{C_i}{\sum_k C_k} P \quad (31)$$

where, $\sum_k C_k = C_{H_2} + C_{C_2H_2} + C_{C_2H_4} + C_{C_2H_6} + C_{C_4H_8} + C_{C_6H_{12}} + C_{CO}$.
Differentiating Eq.(31) with respect to z yields:

$$\frac{dp_i}{dz} = \frac{P}{\sum_k C_k} \frac{dC_i}{dz} \quad (32)$$

From Eq.(26) and Eq.(32), we get:

$$\frac{u_g \sum_k C_k}{P \rho_s} \frac{dp_i}{dz} = \sum_j r_{(j,i)} \quad (33)$$

The resultant ODEs for component mass balance in partial pressure terms are:

$$\frac{dp_i}{dz} = \frac{P \rho_s}{u_g \sum_k C_k} (r_{1,i} + r_{2,i} + r_{3,i} + r_{4,i}) \quad (34)$$

Hence, the ODEs for component mass balance in functional form can be represented by:

$$\begin{aligned} \frac{dp_i(z)}{dz} = f \left(p_{CO}(z), p_{C_2H_2}(z), p_{H_2}(z), T(z), \right. \\ \left. p_{C_2H_4}(z), p_{C_2H_6}(z), p_{C_4H_8}(z), p_{C_6H_{12}}(z) \right) \end{aligned} \quad (35)$$

The ODE for energy balance along the length of the reactor is given by:

$$u_g \rho_g c_p \frac{dT}{dz} = \sum_j (-\Delta H_{(r,j)}) \rho_s r_j \quad (36)$$

The resultant ODE for energy balance in functional form is:

$$\begin{aligned} \frac{dT}{dz} = g \left(p_{CO}(z), p_{C_2H_2}(z), p_{H_2}(z), T(z), p_{C_2H_4}(z), \right. \\ \left. p_{C_2H_6}(z), p_{C_4H_8}(z), p_{C_6H_{12}}(z) \right) \end{aligned} \quad (37)$$

This work is more focused on root cause analysis for slippage events. Hence, the ODE for the component mass balance of (C_2H_2) is considered:

$$\begin{aligned} \frac{dp_{C_2H_2}(z)}{dz} = f \left(p_{CO}(z), p_{C_2H_2}(z), p_{H_2}(z), T(z), \right. \\ \left. p_{C_2H_4}(z), p_{C_2H_6}(z), p_{C_4H_8}(z), p_{C_6H_{12}}(z) \right) \end{aligned} \quad (38)$$

Discretizing this ODE using the forward difference method leads to:

$$\begin{aligned} \frac{p_{C_2H_2}(z+1) - p_{C_2H_2}(z)}{\Delta z} = f \left(p_{CO}(z), p_{C_2H_2}(z), p_{H_2}(z), T(z), \right. \\ \left. p_{C_2H_4}(z), p_{C_2H_6}(z), p_{C_4H_8}(z), p_{C_6H_{12}}(z) \right) \end{aligned} \quad (39)$$

After, algebraic manipulation, we can rewrite Eq. (39) as (assuming $\Delta z = 1$):

$$p_{C_2H_2}(out) = f\left(p_{CO}(in), p_{C_2H_2}(in), p_{H_2}(in), T(in), p_{C_2H_4}(in), p_{C_2H_6}(in), p_{C_4H_8}(in), p_{C_6H_{12}}(in)\right) \quad (40)$$

where *in* and *out* represent the inlet and outlet of the reactor.

With the help of the above equation, a knowledge-based causal digraph for the slippage event has been shown in Fig. 22. To obtained the causal relation between the tags we considered the tags in partial pressure unit. Here, $p_{CO}(in)$, $p_{C_2H_2}(in)$, $p_{H_2}(in)$, $T(in)$, $p_{C_2H_4}(in)$ are the measured tags. On the other hand $p_{C_2H_6}(in)$, $p_{C_4H_8}(in)$, $p_{C_6H_{12}}(in)$ are the unmeasured tags. All these measured and unmeasured tags affect $P_{C_2H_2}$ out of the reactor. This is shown in Fig. 59.

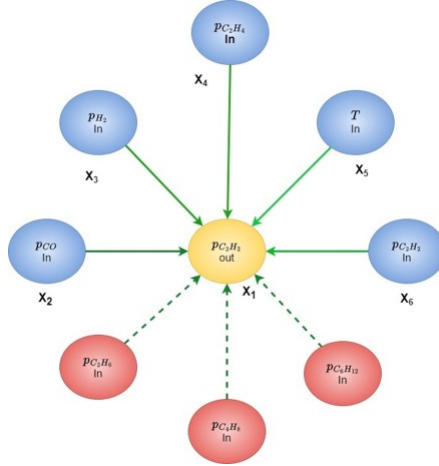


Figure 59: Acetylene process: digraph from first principal model

For subsequent analysis, only the six (measured) tags (shown in blue colour in Fig. 59) have been considered for the slippage event. For ease of convenience, we label these six tags from $X_1, \dots, X_6$ as shown in Table 7.

Variable	Tags
X_1	C_2H_2 Out
X_2	CO in
X_3	H_2 in
X_4	C_2H_4 in
X_5	T_{in}
X_6	C_2H_2 in

Table 7: Variables corresponding to tags

Before applying the T^2 approach, the variables are normalized as:

$$Z_i = \frac{X_i - \mu_i}{\sigma_i}, \quad i = 1, 2, \dots, 6 \quad (41)$$

Here, μ_i and σ_i are the mean and standard deviation respectively of variable X_i . The mean and standard deviation should be computed for the nominal data where there is no slippage event or fault. The T_{Data}^2 value is computed using the following equation.

$$T_{Data}^2 = Z^T \Sigma^{-1} Z \quad (42)$$

where, Σ is correlation matrix of considered data. The spike in T_{Data}^2 value indicates the occurrence of a fault which could be a slippage event. As discussed in Section 3.3, for root cause analysis, we need to obtain T^2 decomposition for the six tags based on the digraph presented in Fig. 59. The corresponding T^2 decomposition is given below:

$$T_{Model}^2 = T_{X_1|(X_2, X_3, X_4, X_5, X_6)}^2 + T_{X_2}^2 + T_{X_3}^2 + T_{X_4}^2 + T_{X_5}^2 + T_{X_6}^2 \quad (43)$$

Here,

$$T_{X_2}^2 = Z_2^2 \quad (44)$$

$$T_{X_3}^2 = Z_2^3 \quad (45)$$

$$T_{X_4}^2 = Z_2^4 \quad (46)$$

$$T_{X_5}^2 = Z_2^5 \quad (47)$$

$$T_{X_6}^2 = Z_2^6 \quad (48)$$

The $T_{X_1|(X_2, X_3, X_4, X_5, X_6)}^2$ quantity is computed using conditional probability approach as discussed in Section 3.3. In particular, $T_{X|Y}^2$ is given as follows:

$$T_{X|Y}^2 = \frac{(Z_X - \mu_{X|Y})^2}{\Sigma_{X|Y}} \quad (49)$$

where, $X = X_1$ and $Y = (X_2, X_3, X_4, X_5, X_6)$. Also, $\Sigma_{X|Y} = \Sigma_{XX} - \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YX}$ and $\mu_{X|Y} = \Sigma_{XY} \Sigma_{YY}^{-1} Z_Y$ are the conditional mean and variance of normalized X_1 i.e. of Z_1 given variables $X_2, \dots, X_6$. Here, $Z_Y = [Z_2, Z_3, Z_4, Z_5, Z_6]^T$.

In this study, the root cause analysis of slippage events is carried out both with hourly data, as well as minute-wise data. It is expected that once comprehensive analysis is performed in future, the minute-wise data will yield better performance in terms of root cause identification since many phenomena will not be represented in hourly data. We now present a few specific cases for these analyses.

A.2 Root cause analysis of slippage with hourly data

The root cause analysis of slippage with hourly data is performed for two cases: one where no slippage event occurs and another where a slippage event is present.

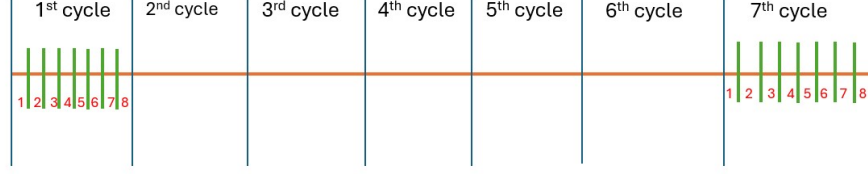


Figure 60: Cycle-wise segmentation of Reactor A Data

By comparing these scenarios, it can be evaluated how the T^2 decomposition method performs under different conditions. This provides insights into the effectiveness of method in detecting the slippage and identifying the root cause.

To deal with the non-stationary nature of data, cycle-1 and cycle-7 for Reactor A are divided into eight windows of length 500 data points each as shown in Fig. 60. In this analysis, 1st window (14th February 2017, 03 : 00 to 6th March 2017, 22 : 00) of cycle-1, 4th window (22nd January 2024, 23 : 00 to 12th February 2024, 18 : 00) of cycle-7, and 7th window (15th April 2024, 07 : 00 to 6th May 2024, 02 : 00) of cycle-7 are considered.

A.2.1 Without Slippage Event

For this case, we analyze Reactor A during the first window of cycle-1, using 500 data points for T^2 decomposition. The T^2_{Data} , T^2 decomposition and corresponding raw data plots are examined as shown in Fig. 61b, Fig. 62, and Fig. 63 respectively.

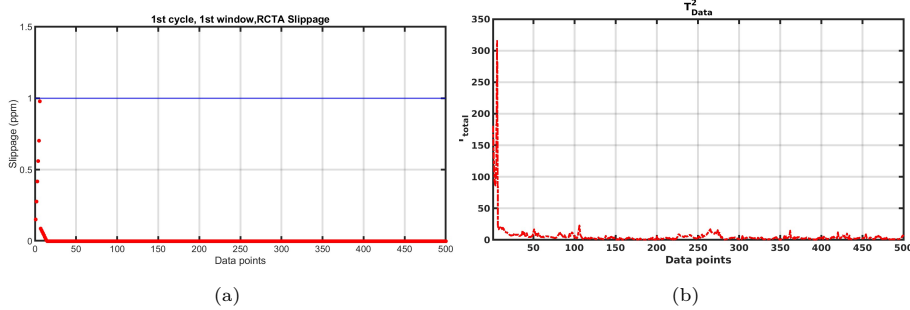


Figure 61: Cycle 1 window 1: (a) Slippage plot (b) T^2_{Data} plot

We observe that there is no slippage event as shown in Fig. 61a. Correspondingly, there is no significant spike in T^2_{Data} as shown in Fig. 61b. The initial spike in T^2_{Data} is caused by non-zero slippage. However, for a slippage event to occur, the concentration of C_2H_2 out (acetylene) must exceed 1 ppm. Similarly, there are no spikes in the decomposed T^2 values as depicted in Fig. 62. Ad-

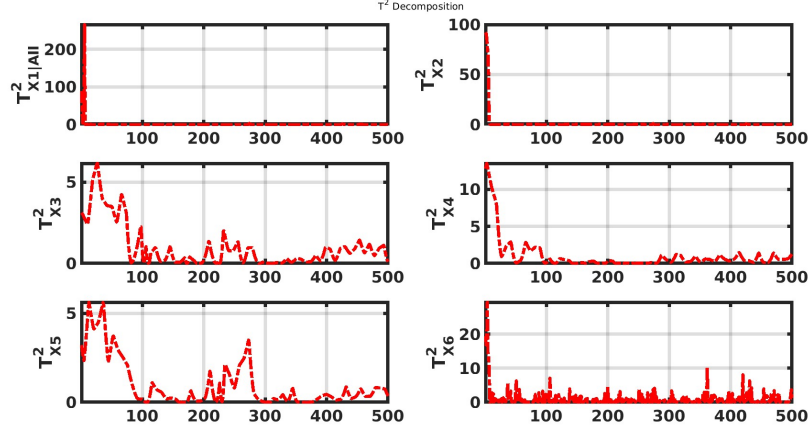


Figure 62: Cycle 1 window 1: T^2 Decomposition plot

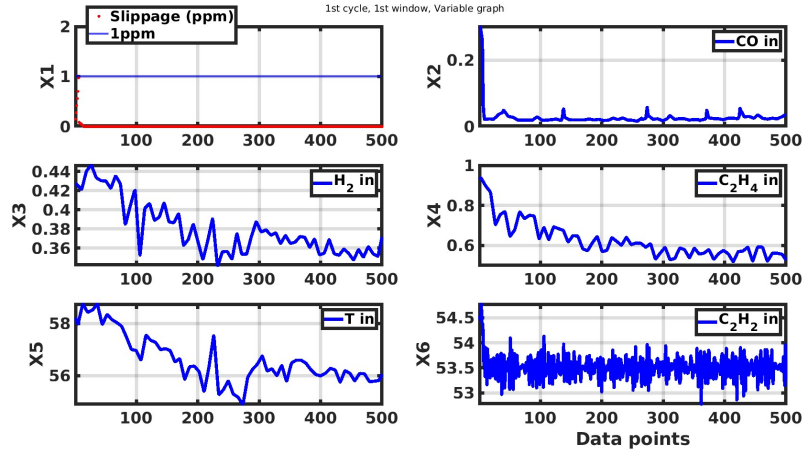


Figure 63: Cycle 1 window 1: Raw data plot

ditionally, the raw data presented in Fig. 63 appears to be normal throughout the window. This normal behavior in T^2 indicates that the reactor is operating as expected without any slippage.

A.2.2 With Slippage Event

In this study, 7th cycle of Reactor A is considered. For this cycle, we considered two cases: (i) Analysis of window 7 data and (ii) Analysis of window 4 data.

Case 1: Analysis of window 7 data

In this case, there are total 14 slippage events out of 381 data points as shown in Fig. 64. For root cause analysis we considered three slippage events at instant 207, 209, and 210 respectively.

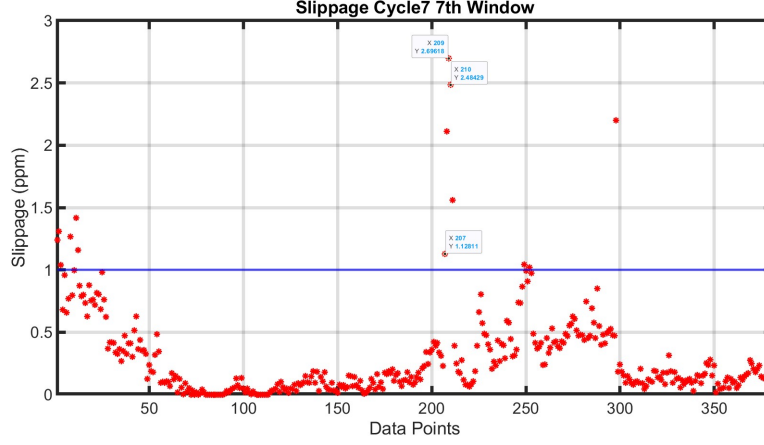


Figure 64: Cycle 7 window 7: Slippage event plot

Using Equation 42, T_{Data}^2 is computed and the corresponding plot is shown in Fig. 65. As illustrated in the figure, there are significant spikes at the same instants where slippage events were identified (instants 207, 209, and 210). These spikes indicate that faults likely occurred at these positions. After identifying

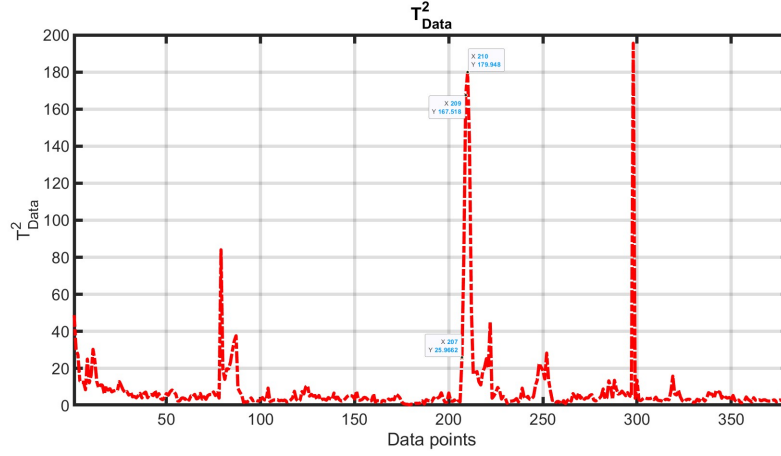


Figure 65: Cycle 7 window 7: T_{Data}^2 plot

the significant spikes in T_{Data}^2 the next step is to determine the root cause of these anomalies. To do this, we need to investigate which specific variables are

contributing to the increase in T_{Data}^2 . By analyzing the tag-wise T^2 decomposition, we can pinpoint which variables are root causes leading to the significant spikes in T_{Data}^2 . To proceed, decomposed T^2 value for each tag is analyzed. This approach allows us to isolate the contributions of each variable and identify the particular ones that are driving the spikes. Through this detailed examination, we aim to uncover the underlying issues responsible for the observed faults and better understand the nature of the deviations in the data. The various decomposed T^2 Decomposition as listed in Eqn. 43 are plotted in Fig. 66 for the six variable discussed earlier.

From Fig. 66, we can see that $T_{X_1|All}^2$ (top panel, first column) shows signifi-

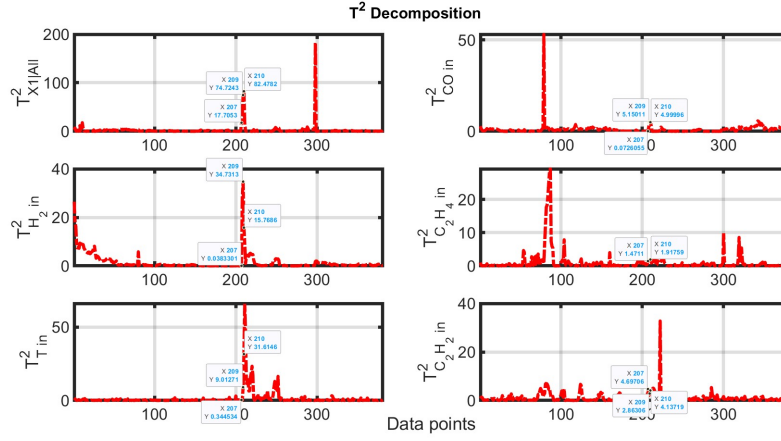


Figure 66: Cycle 7 window 7: T^2 Decomposition plot

cant spikes at instant 209 and 210 from its mean. At 209 instant, the $T_{X_1|All}^2$, $T_{X_3}^2$ and $T_{X_5}^2$ exhibit spikes. This suggests that a potential root cause of the slippage at instant 209 could be related to the X_3 , X_5 and unmeasured variables. Therefore, we examine the raw data at this instant to better understand the underlying phenomena. In Fig. 67, the raw data for all the six variables has been plotted. For this figure, we can see that variable X_3 which is H_2 inlet to the reactor has significantly dropped at instant 209. This makes sense since if the inlet flow of H_2 to the reactor is reduced, the rate of desired reaction to produce C_2H_4 (the desired product) will decrease and C_2H_2 will exit the reactor in larger quantities, leading to a slippage event. Additionally, we also observe that variable X_5 , which represents the inlet temperature T_{in} , increases simultaneously. This might be due to operator intervention; by raising the inlet temperature, the desired reaction rate can be enhanced, which reduces the amount of slippage.

Case 2: Analysis of window 4 data

In this case, there are total of 23 slippage events out of 500 data points as shown in Fig. 68. The slippage events are reflected by spikes in T_{Data}^2 as shown in

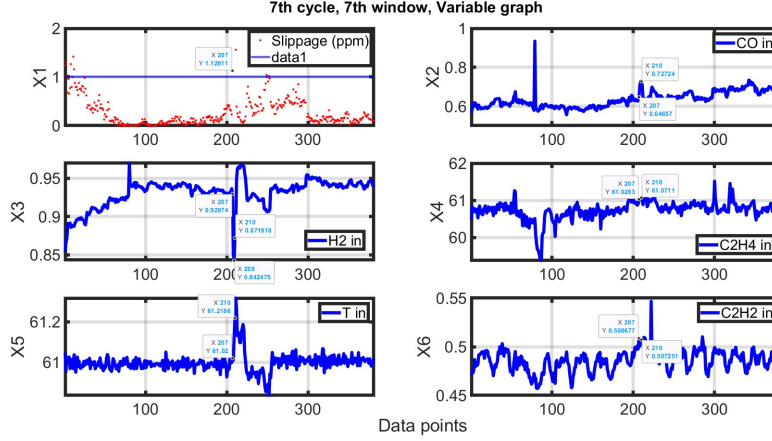


Figure 67: Cycle 7 window 7: Raw tag plot

Fig. 69. Also, there are significant spikes in $T_{X_1|All}^2$ as shown in Fig. 70. This indicates that the root cause could be variables other than the five variables considered as potential causes for the slippage in the current analysis. The data being analysed is hourly averaged data that may not accurately reflect the process dynamics and the instantaneous control action taken by the controller or the operator. For example, from Fig. 71, it can be observed from plot of X3 (H_2 in) that from instant 1 to 100 (23^{rd} January 2024, 01:00 to 27^{th} January 2024, 05:00), there is an increasing trend in this variable (see circled portion). However its effect on slippage cannot be teased out using averaged-hourly data. Therefore in subsequent section, we consider analysis of minute-wise data from 23^{rd} January 2024, 21:18 to 25^{th} January 2024, 23:17, which corresponds to the same 100 data points in the hourly dataset.

A.3 Root cause analysis with minute-wise Data

In this analysis, minute-wise data is considered. The minute-wise data shared with us is of reactor A, 7^{th} cycle from 21^{st} November 2023, 00:00 to 29^{th} February 2024, 22:59, i.e. a total of 1,45,380 data points. Out of these data points, 3000 data points from instant 92,000 to 94,999 (23^{rd} January 2024, 21:18 to 25^{th} January 2024, 23:17) are considered for current analysis. There are 296 slippage events in this period, out of which two are explained in this study as highlighted in Fig. 72. It can be observed from Fig. 72 that there is slippage at 450 and 1438 instants. Correspondingly significant spikes are observed in T_{Data}^2 plot (Fig. 73). The T^2 decomposition plot in Fig. 74 shows that there is significant spike at 450^{th} instant in $T_{C_2H_2|All}^2$ and $T_{H_2}^2$ in which implies that H_2 in is surely the root cause along with other variables which are not measured. Additionally, the root cause of the slippage event at the 1438^{th} instance is attributed to variables that were not measured. From Figs. 75 and 76 it can

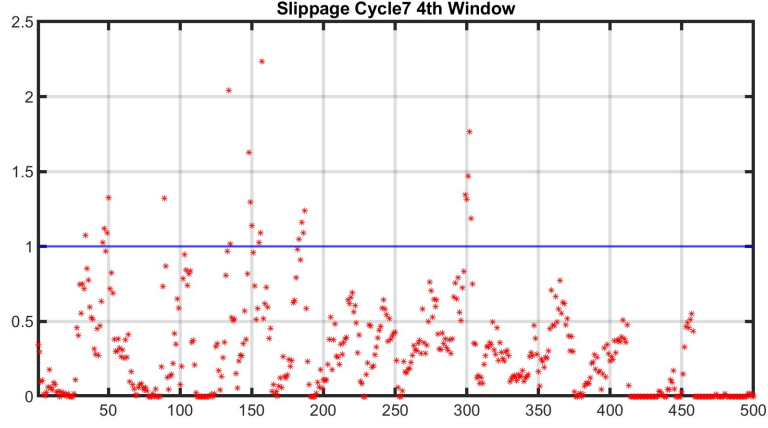


Figure 68: Cycle 7 window 4: Slippage event plot

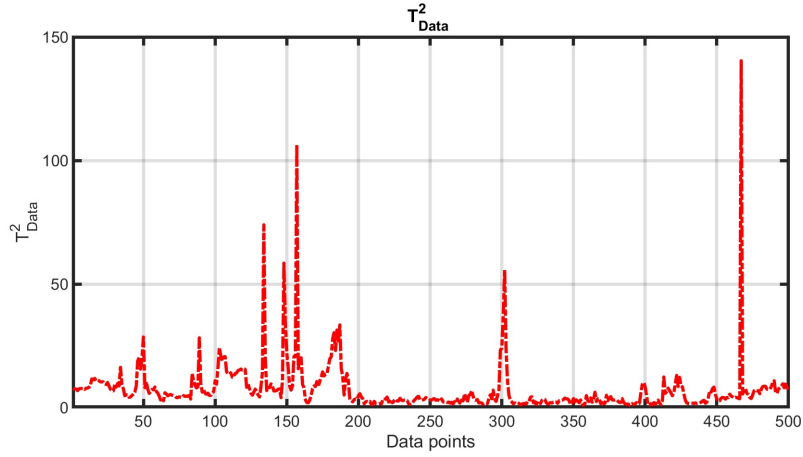


Figure 69: Cycle 7 window 4: T^2_{Data} plot

be deduced that the slippage occurred because the value of H_2 in and value of H_2/C_2H_2 mole ratio has reduced. For the slippage event at 1438 instant from Fig. 74 it is clear that other than in $T^2_{C_2H_2|All}$ there is no significant spike in T^2 of other variables. This explains that there is slippage at 1438 instant but the variables considered in the analysis are not its root cause. This argument is also supported by Fig. 75 in which it can be observed that there is no significant variation in any of the variables around this instant. Therefore, there might be other variable(s) responsible for slippage event which are not considered in the current analysis. Note that from Figure 75 it can be conjectured that when slippage occurred at 1438 instant, the operator or controller increased the H_2 in

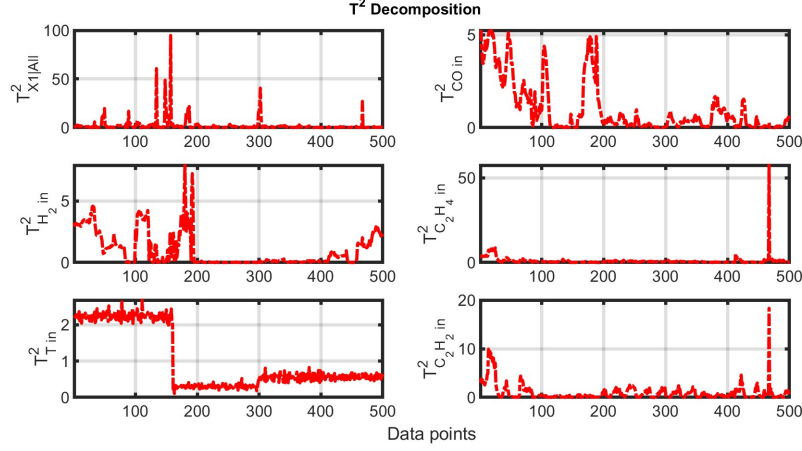


Figure 70: Cycle 7 window 4: T^2 Decomposition plot

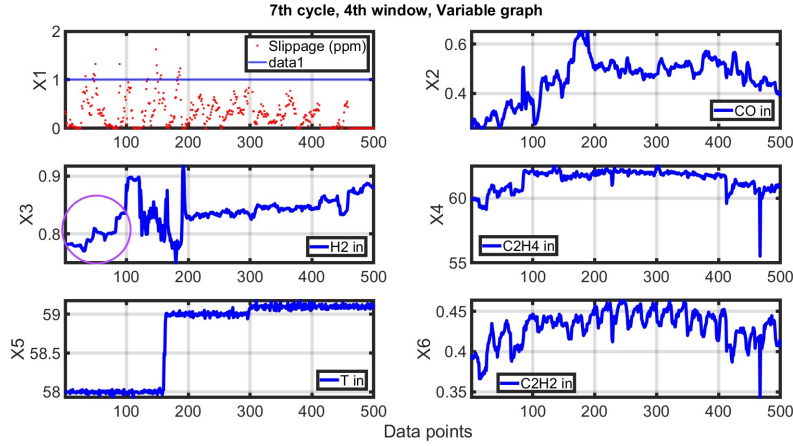


Figure 71: Cycle 7 window 4: Raw tag plot

to counter the slippage.

From these two implementations of T^2 decomposition on hourly and minute-wise data, it is clear that the adopted method has potential to effectively detect the slippage and identify its root cause. In cases where T^2 decomposition using hourly data failed to explain the root cause, analyzing minute-wise data provided deeper insights. Slippage events for which root cause could not be identified point to the need to include more variables for digraph generation and subsequent analysis.

It should be noted that T^2_{Data} can be high even if the slippage has not occurred. This is seen in Figure 69 where at about 470th time instant, the

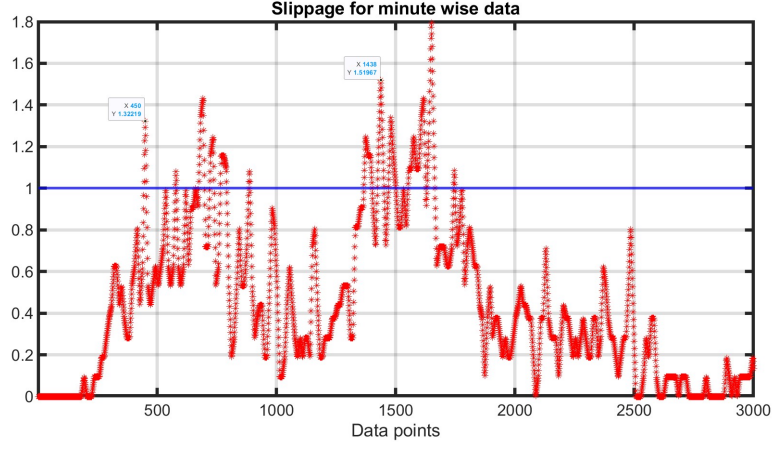


Figure 72: Cycle 7 minute wise: Slippage event plot

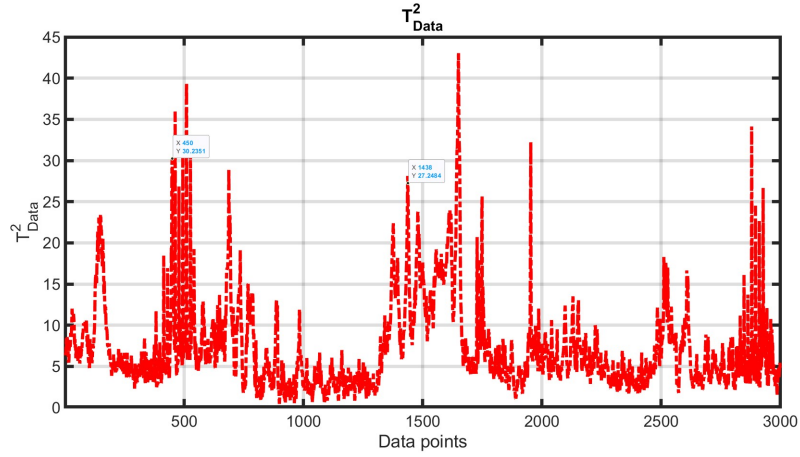


Figure 73: Cycle 7 minute wise: T^2_{Data} plot

T^2_{Data} is very high but slippage has not occurred. The T^2_{Data} increase simply tells that something different is happening in the process compared to the recent operation. It may or may not correspond to a slippage event. If the user wants to focus only on slippage then this T^2_{Data} based trigger can be ignored. Instead the T^2 decomposition can be directly evaluated to identify the root cause whenever slippage occurs.

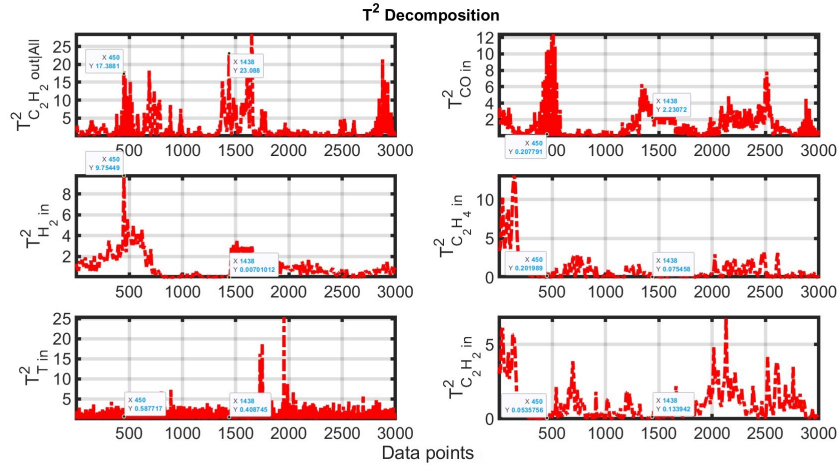


Figure 74: Cycle 7 minute wise: T^2 Decomposition plot

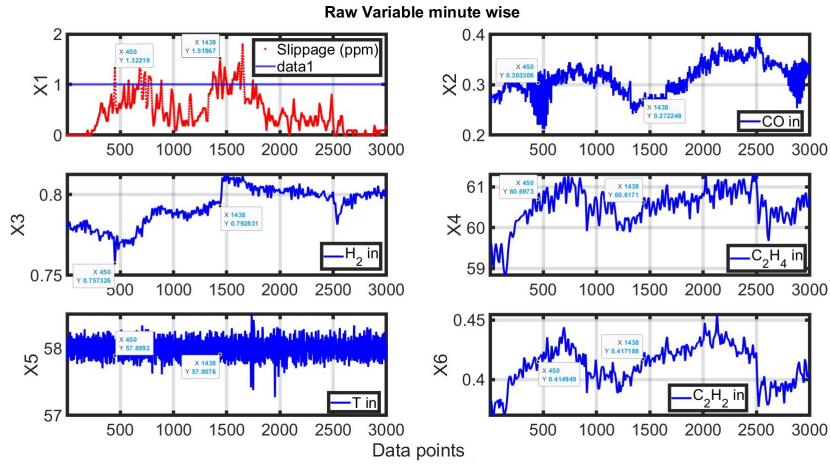


Figure 75: Cycle 7 minute wise: Raw tag plot

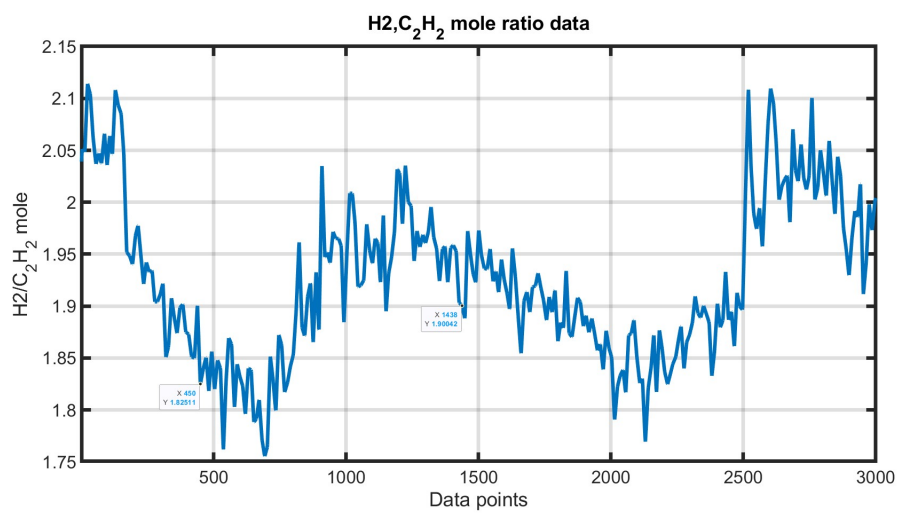


Figure 76: Cycle 7 minute wise: H_2/C_2H_2 mole ratio

B Appendix: Digraph generation, T^2 computation and Root cause analysis of slippage event for without transformed data

B.1 Digraph generation from without transformed data

The raw data obtained after the data pre-processing for the considered 14 variables is further used to develop a digraph using the PC algorithm. Fig. 77 shows the obtained digraph. Based on obtained causal network or digraph, the decomposition of T^2 statistics is done as explained in section 3.3 with without transformed data. To enhance the readability of the graph, each node is assigned a new name, denoted as X_i , where i ranges from 1 to 14, corresponding to the original tag names. It is important to note that, while developing the digraph, a specific approach is used to remove the absurd edges that do not align with fundamental principles. In this approach, a restriction is imposed to prevent edges between certain nodes among the input variables, intermediate variables, and output variables. Table 4 lists the variables grouped into input, intermediate, and output categories. Figure 44 illustrates the imposed restrictions. The nodes between which no edges are allowed are shown, with restrictions applied to certain connections among the input, intermediate, and output variables. A two-sided arrow indicates no restriction, allowing the PC algorithm to determine the presence and direction of the edge.

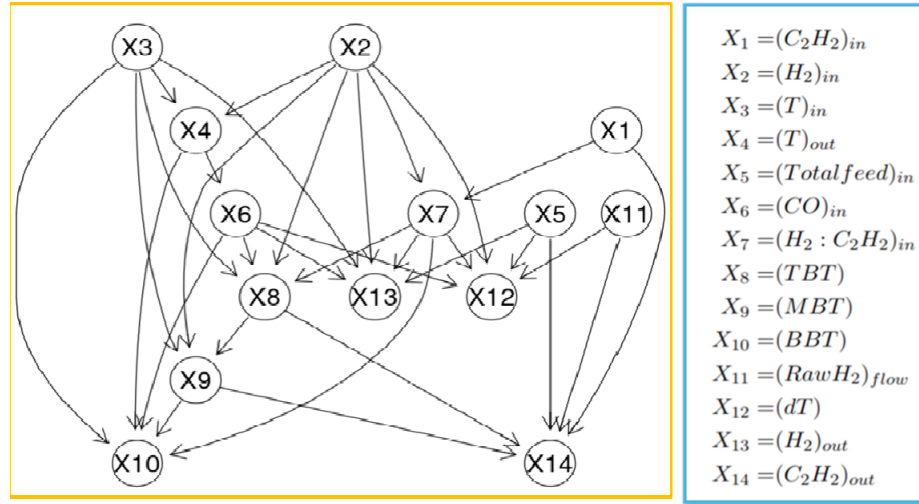


Figure 77: Digraph using without transformed data with considered 14 variables

B.2 Window wise T^2 computation from without transformed data

Based on the causal network or digraph discussed in the previous subsection, the T^2 statistic is decomposed as outlined in Section 3.3, using without transformed data. The T^2 computation requires the first and second moments of the data. To process the data efficiently, a sliding window approach is employed, where each window has a length of $50n$, with n representing the number of tags considered. This parameter n is a tuning factor, chosen heuristically. The window slides one sample at each sampling interval. The expression for the decomposition of T^2 statistics based on the obtained digraph and considered 14 variables is given as follows:

$$\begin{aligned} T^2 = & T_{X_1}^2 + T_{X_2}^2 + T_{X_3}^2 + T_{X_4|(X_2, X_3)}^2 + T_{X_5}^2 + T_{X_6|X_4}^2 + T_{X_7|(X_1, X_2)}^2 + \\ & T_{X_8|(X_2, X_3, X_6, X_7)}^2 + T_{X_9|(X_2, X_3, X_8)}^2 + T_{X_{10}|(X_3, X_4, X_6, X_7, X_9)}^2 + T_{X_{11}}^2 + \\ & T_{X_{12}|(X_2, X_5, X_6, X_7, X_{11})}^2 + T_{X_{13}|(X_2, X_3, X_5, X_6, X_7)}^2 + \\ & T_{X_{14}|(X_1, X_5, X_8, X_9, X_{11})}^2 \end{aligned}$$

The threshold value for each of the 14 decomposed T^2 of variables is computed using the expression discussed in subsection 3.4. The ultimate objective of this project is to identify the root cause of the slippage events. A thorough analysis of the provided acetylene reactor dataset reveals that when slippage occurs, it persists for a significant duration. As a result, multiple slippage events can be viewed as part of a single episode, allowing the entire dataset to be decomposed into multiple episodes for root cause analysis. The root cause analysis of one such episode is discussed in the next subsection.

B.3 Episode-wise root cause analysis of slippage event using without transformed data

The episode selected for root cause analysis spans from the 950th to the 1050th time instant (in minutes). The first slippage event in this episode occurred at the 1026th instant (on December 12, 2023, at 19:03), and the final slippage event took place at the 1041st instant (on December 12, 2023, at 19:17). To investigate the root cause of the slippage event, a specific approach is employed in which we analyze the T^2 decomposition values of each variable from 30 to 35 time instants before the occurrence of the first slippage event in the episode. Fig. 78 to Fig. 82 shows plots of Raw data (left) and T^2 decomposition (Right) vs time instant for transformed data. The blue continuous line in the right side plots are values of T^2 at every instant and instances with pink dots depict slippage events. The red constant line is for the computed threshold value. Inspecting T^2 decomposition values of each variable in Fig. 78 to Fig. 82, it can be observed that before the first slippage occurred at 1026th instant in this episode, T^2 decomposition values of C_2H_2 In, T In, *Total Feed*, *CO In* and

Raw H_2 has crossed their corresponding threshold whereas T^2 decomposition values of other variables are within the threshold limit. This indicates that the above variables are potential root causes of slippage events in this episode. The raw data for the above variables show significant deviation from their mean value when the T^2 decomposition values are above the threshold which confirms that the above variables are the potential culprit. Fig. 83 summarises the above analysis for this episode.

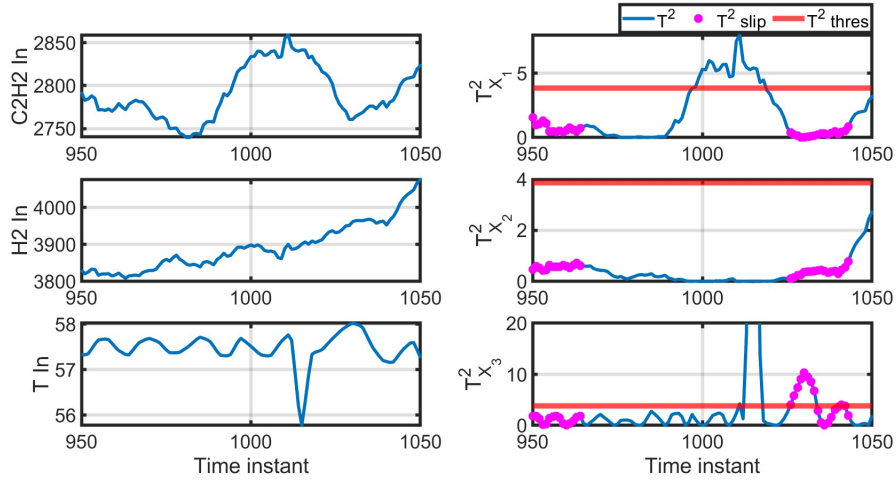


Figure 78: Plot of Raw data (left) and T^2 decomposition (Right) vs time instant for without transformed data: (Top) C_2H_2 in, (Mid) H_2 in, and (Bottom) T in

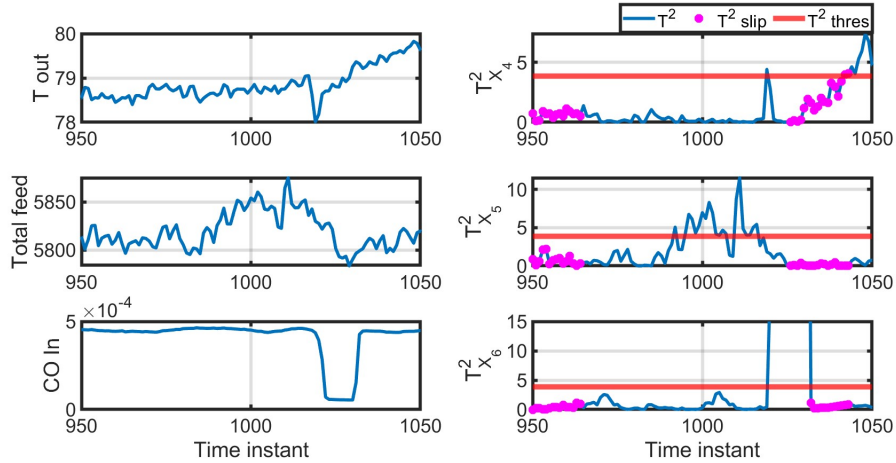


Figure 79: Plot of Raw data (left) and T^2 decomposition (Right) vs time instant without transformed data: (Top) T out, (Mid) Total feed, and (Bottom) CO in

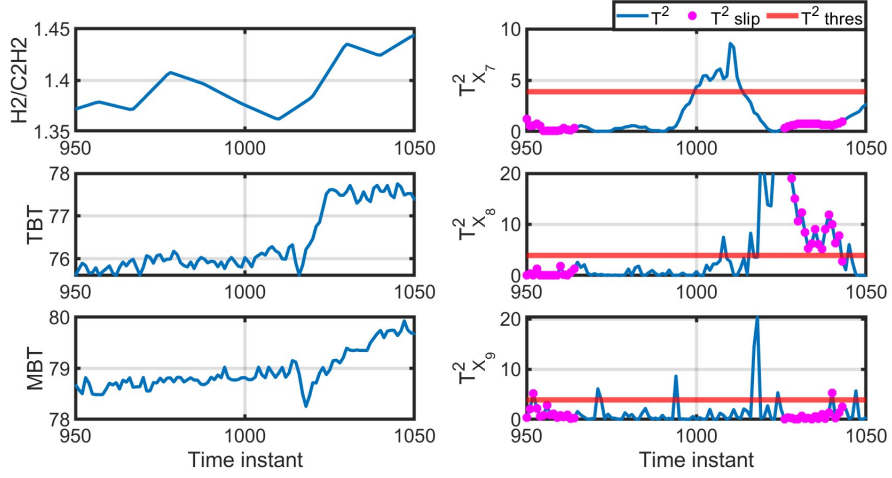


Figure 80: Plot of Raw data (left) and T^2 decomposition (Right) vs time instant without transformed data: (Top) $H_2 : C_2H_2$ in, (Mid) TBT, and (Bottom) MBT

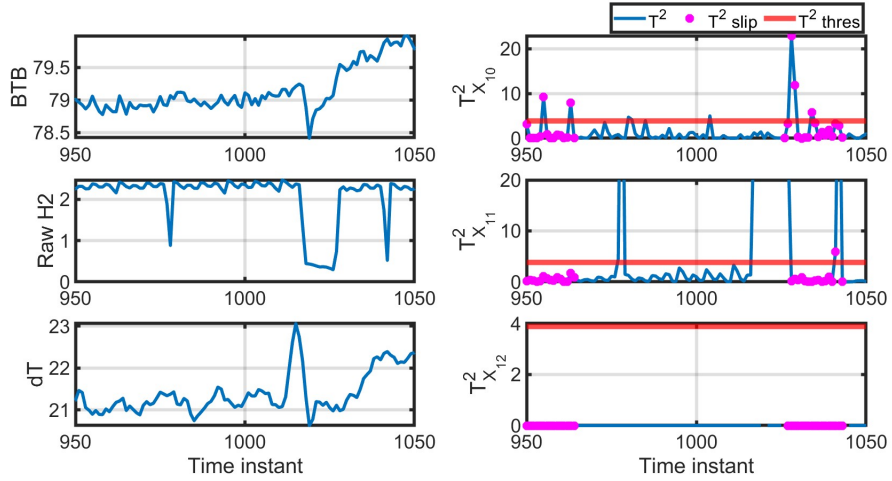


Figure 81: Plot of Raw data (left) and T^2 decomposition (Right) vs time instant without transformed data: (Top) BTB, (Mid) Raw H_2 flow, and (Bottom) dT

Fig. 84 illustrates the abnormal behavior of an individual variable when a slippage event occurs, as the T^2 decomposition value exceeds its threshold in this episode. This plot can be highly beneficial for real-time monitoring of the variables. Fig. 85 displays the frequency with which the T^2 decomposition values of all individual tags exceed their respective thresholds during slippage events, in the absence of faulty measurements. Based on this analysis, some statistical insights into the T^2 decomposition method are presented next.

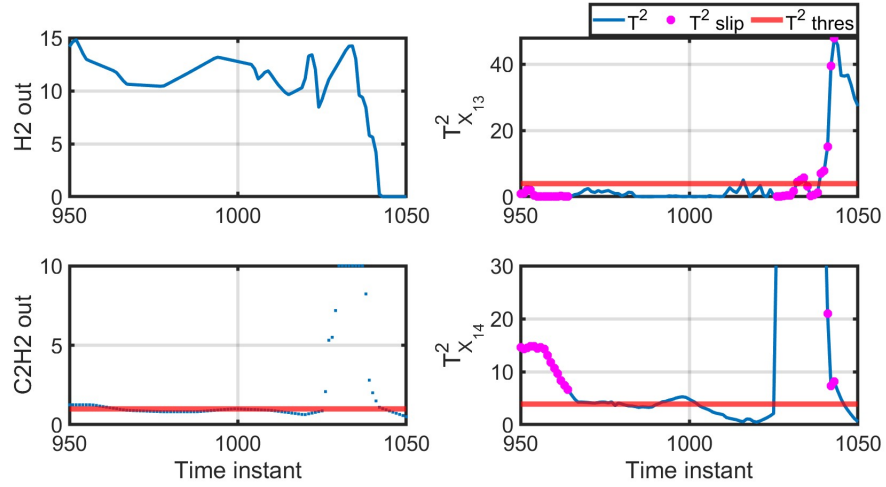


Figure 82: Plot of Raw data (left) and T^2 decomposition (Right) vs time instant without transformed data: (Top) H_2 out, and (Bottom) C_2H_2 out

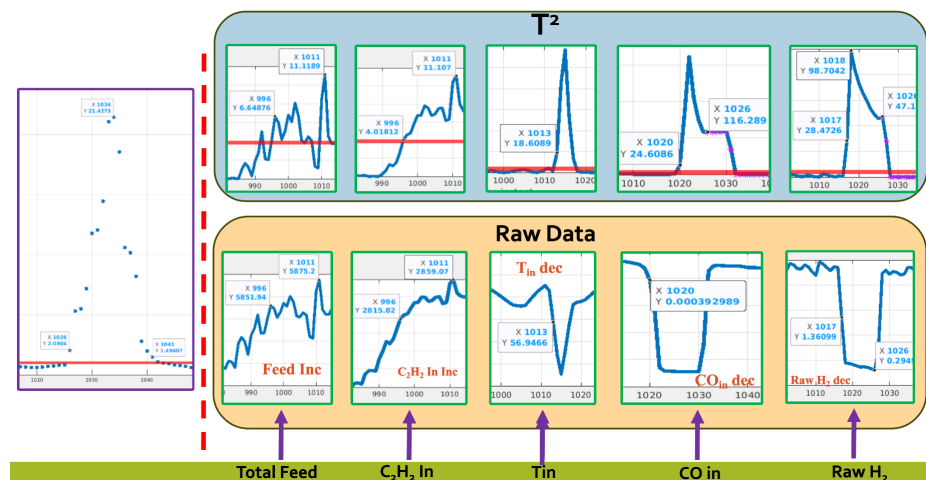


Figure 83: The potential root cause variables for slippage event in the considered episode from without transformed data

B.4 Episode-wise root cause analysis statistics using without transformed data

To assess the effectiveness of the T^2 decomposition methodology used in this work, several metrics are computed using a specific logical approach. Fig. 56 shows that when there are no faulty measurements and a slippage event occurs, the event is considered detected if the T^2 value of any variable exceeds its thresh-

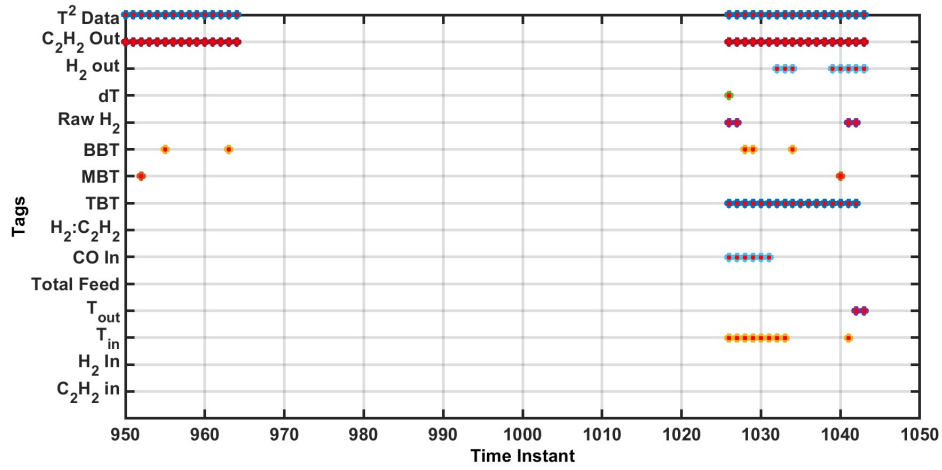


Figure 84: Indicator plot: Real-time indication of abnormal behavior of an individual variable when a slippage event occurs without transformed data.

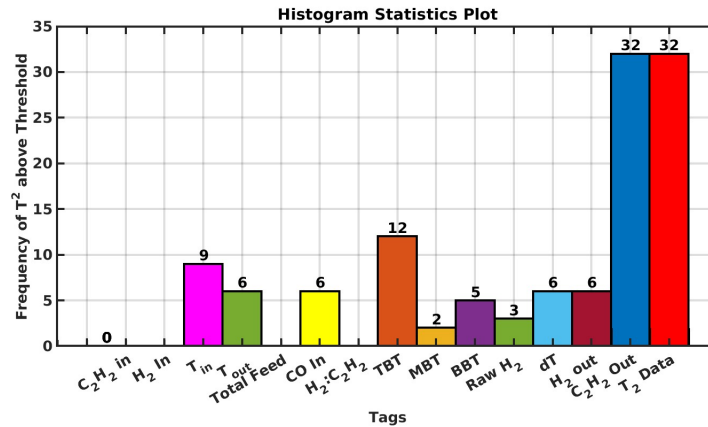


Figure 85: Histogram plot: The number of times T^2 decomposition of all individual tags and T^2 data cross their corresponding threshold for the specified range without transformed data.

old. Fig. 56 illustrates a scenario where, in the absence of faulty measurements, slippage occurs, but the T^2 value of C_2H_2 out does not surpass its threshold, while the T^2 values of other variables do. In this case, the event is regarded as fully explained, as the variable that crosses the threshold is identified as the root cause of the slippage. If none of the other variables cross their thresholds, the event is considered uncaptured.

In the same figure, when slippage occurs and the T^2 value of C_2H_2 out

exceeds its threshold, but no other variables' T^2 values exceed theirs, the event is classified as partially explained. This means that while the variable exceeding the threshold is likely responsible for the slippage, another unmodeled variable may also contribute to the event. If no other variables cross their thresholds, the event is considered unexplained.

Based on these logical rules, the counts for different metrics are summarized in 8 for the episode from the 950th to the 1050th time instant. The table shows that, out of 32 detected slippage events, 0% of the time the method fully explains the root cause, while 62.5% of the time it provides only a partial explanation. This suggests that, in most cases, the root cause of the slippage event is identified during this episode.

Table 8: Episode-wise root cause analysis statistics using without transformed data

Parameter	Statistics
Total number of data points	101
Total Slippage event	32
Total slippage event captured by T2 Data	32 (100%)
Partially explained	20 (62.50 %)
Unexplained	12 (37.50%)
Fully explained	0 (0 %)
Uncaptured case	0 (0 %)

Fig. 86 shows The number of times T^2 decomposition of all individual tags and T^2 data cross their corresponding threshold for the complete data range without transformed data.

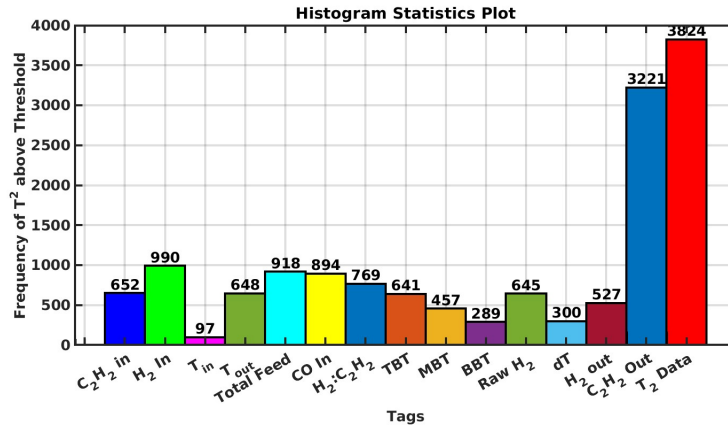


Figure 86: Histogram plot: The number of times T^2 decomposition of all individual tags and T^2 data cross their corresponding threshold for complete data range without transformed data.

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